Digital Design & Computer Arch.

Lecture 8: Instruction Set Architectures II

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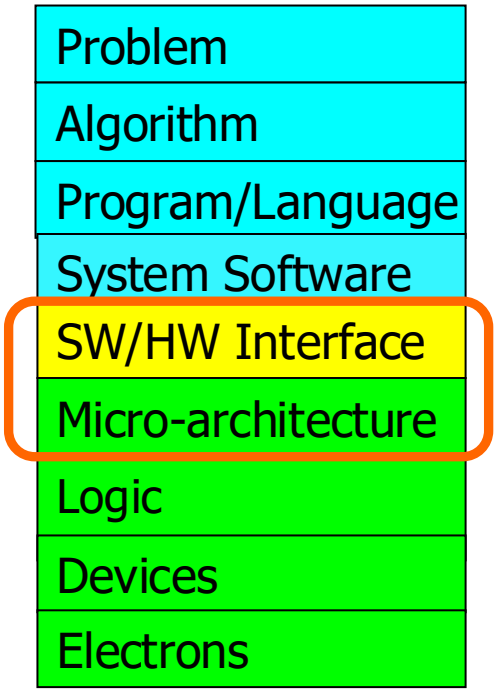
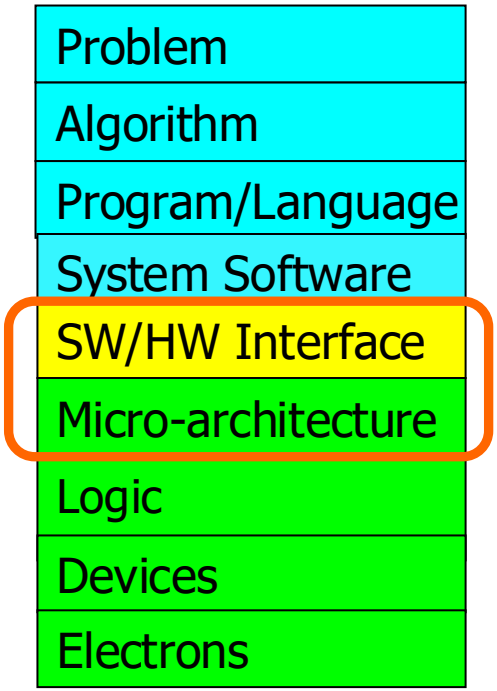
Agenda for Today & Next Few Lectures

- The von Neumann model
- LC-3: An example of von Neumann machine
- LC-3 and MIPS Instruction Set Architectures
- LC-3 and MIPS assembly and programming
- Introduction to microarchitecture and single-cycle microarchitecture
- Multi-cycle microarchitecture



What Will We Learn Today?

- Basic elements of a computer & the von Neumann model
  - LC-3: An example von Neumann machine
- Instruction Set Architectures: LC-3 and MIPS
  - Operate instructions
  - Data movement instructions
  - Control instructions
- Instruction formats
- Addressing modes

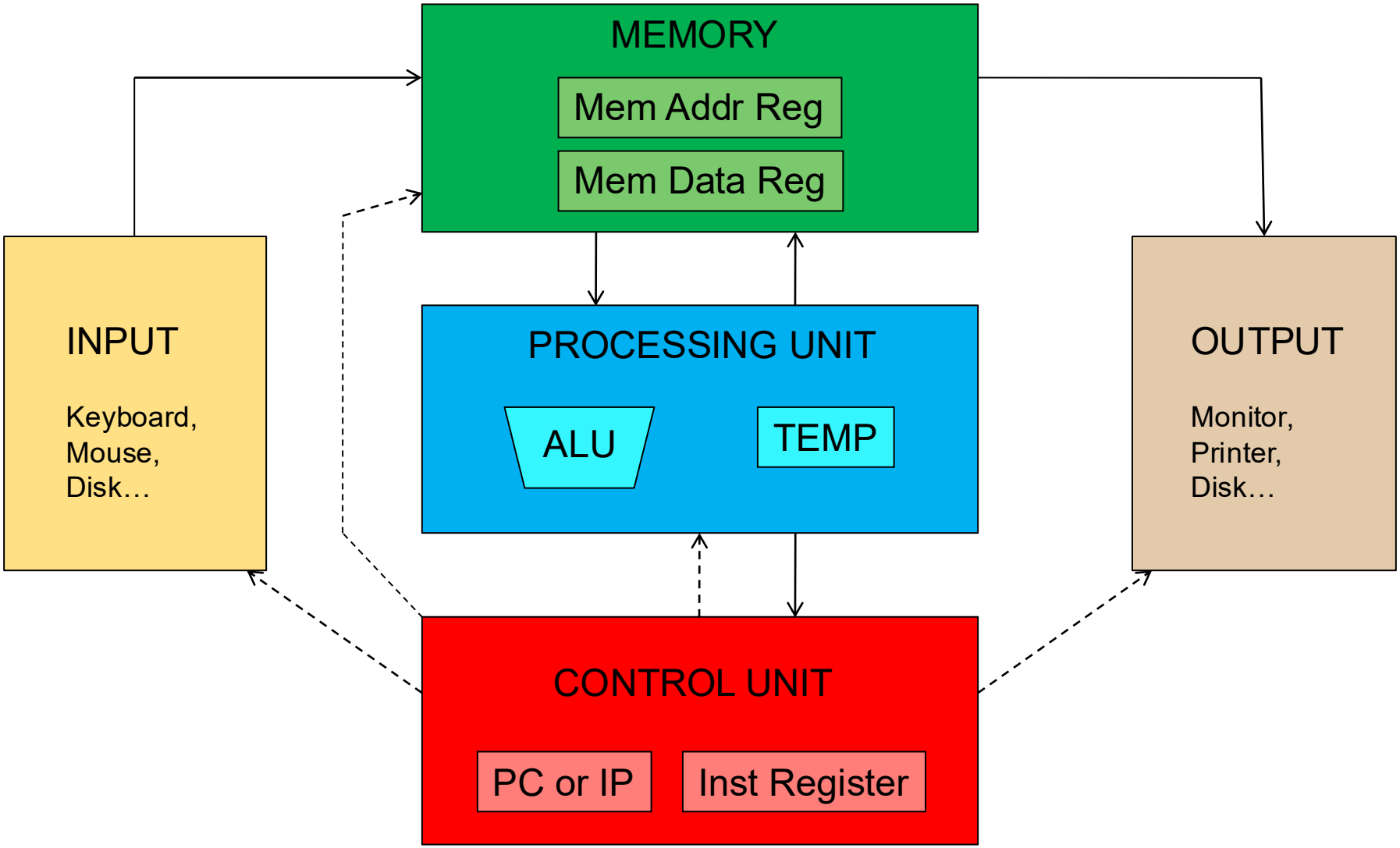


Readings

- This week
  - Von Neumann Model, ISA, LC-3, and MIPS
    - P&P, Chapters 4, 5 (we will follow these today & tomorrow)
    - H&H, Chapter 6 (until 6.5)
    - P&P, Appendices A and C (ISA and microarchitecture of LC-3)
    - H&H, Appendix B (MIPS instructions)
  - Programming
    - P&P, Chapter 6 (we will follow this tomorrow)
  - **Recommended:** H&H Chapter 5, especially 5.1, 5.2, 5.4, 5.5

- Next week
  - Introduction to microarchitecture and single-cycle microarchitecture
    - H&H, Chapter 7.1-7.3
    - P&P, Appendices A and C
  - Multi-cycle microarchitecture
    - H&H, Chapter 7.4
    - P&P, Appendices A and C

Quick Recap: The von Neumann Model



Quick Recap: The von Neumann Model

Stored program

Sequential instruction processing

LC-3: A von Neumann Machine

LC-3: A von Neumann Machine

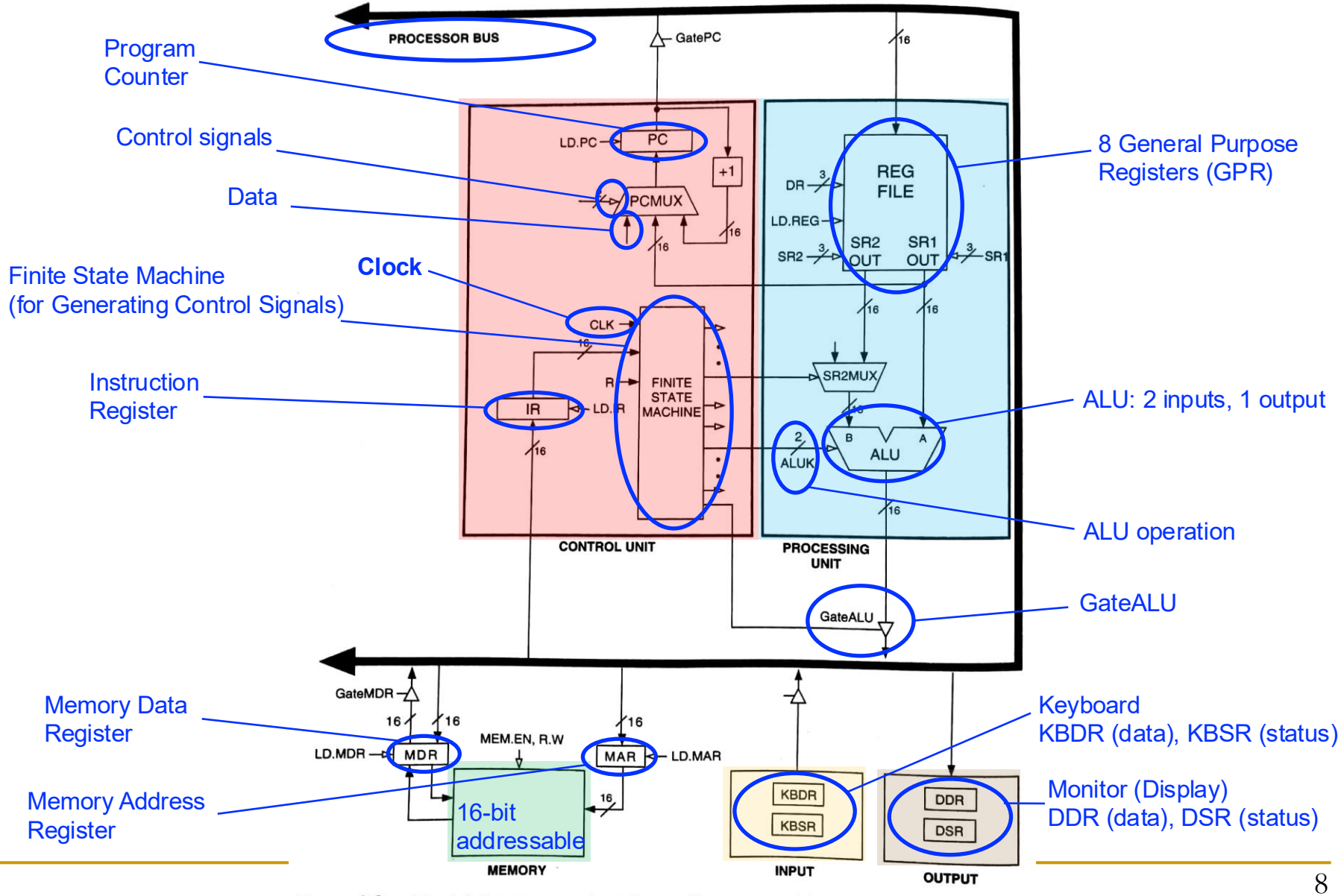


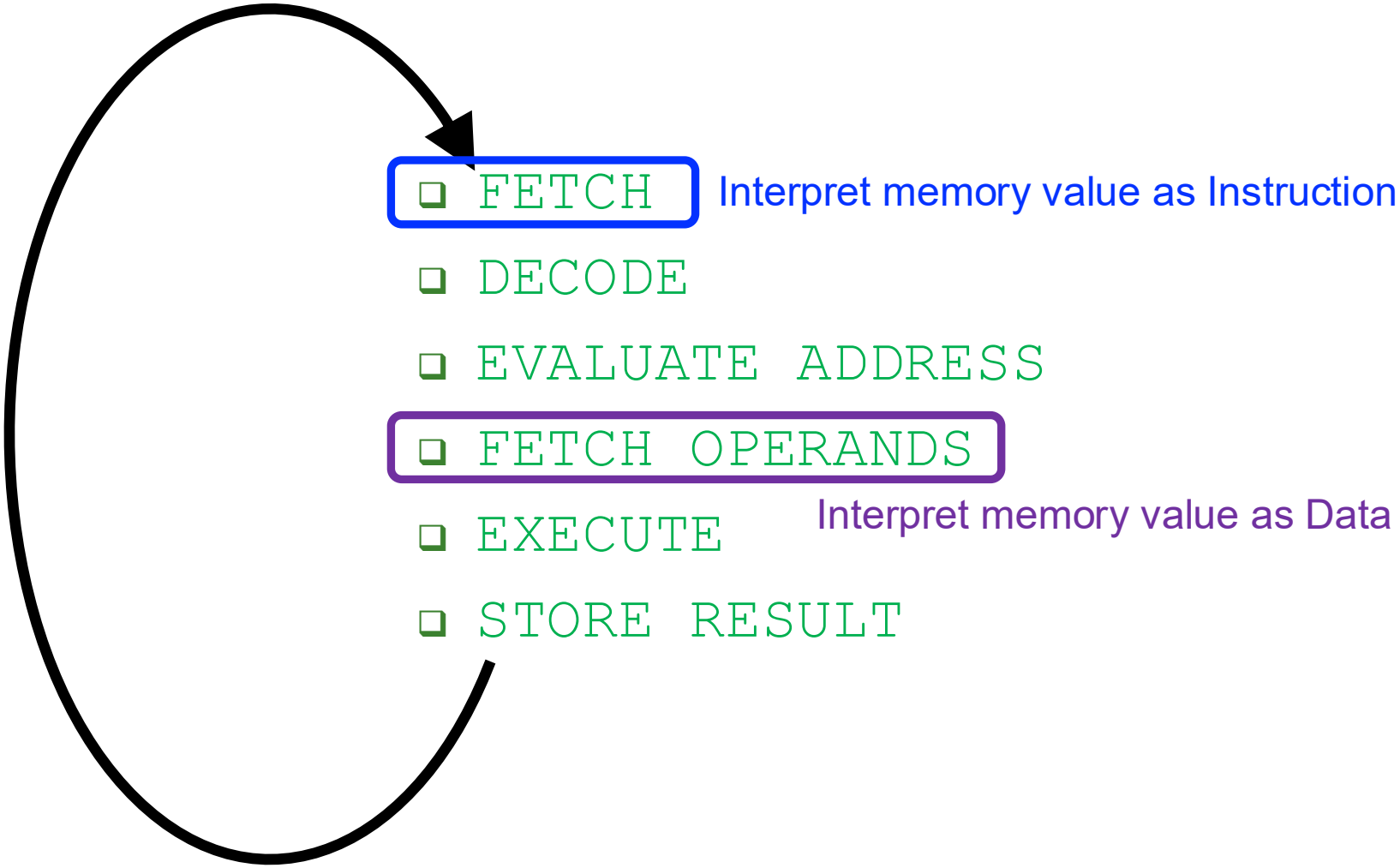
Figure 4.3 The LC-3 as an example of the von Neumann model

Review: Instruction Types

- There are three main types of instructions
- Operate instructions
  - Execute operations in the ALU
- Data movement instructions
  - Read from or write to memory
- Control flow instructions
  - Change the sequence of execution
- Let us start with some example instructions

Instruction (Processing) Cycle

Recall: The Instruction Cycle



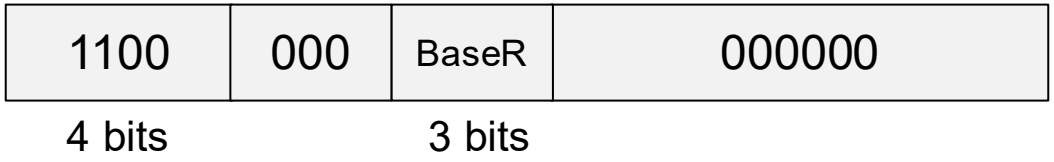
Whether a value fetched from memory is interpreted as an instruction depends on **when** that value is **fetched** in the instruction processing cycle

Recall: Jump in LC-3

- Unconditional branch or jump

- LC-3

```
JMP   R2
```



  - BaseR = Base register
  - PC ← R2 (Register identified by BaseR)

This is register addressing mode

  - Variations
    - RET: special case of JMP where BaseR = R7
    - JSR, JSRR: jump to subroutine

PC UPDATE in LC-3

JMP loads
SR1 into PC

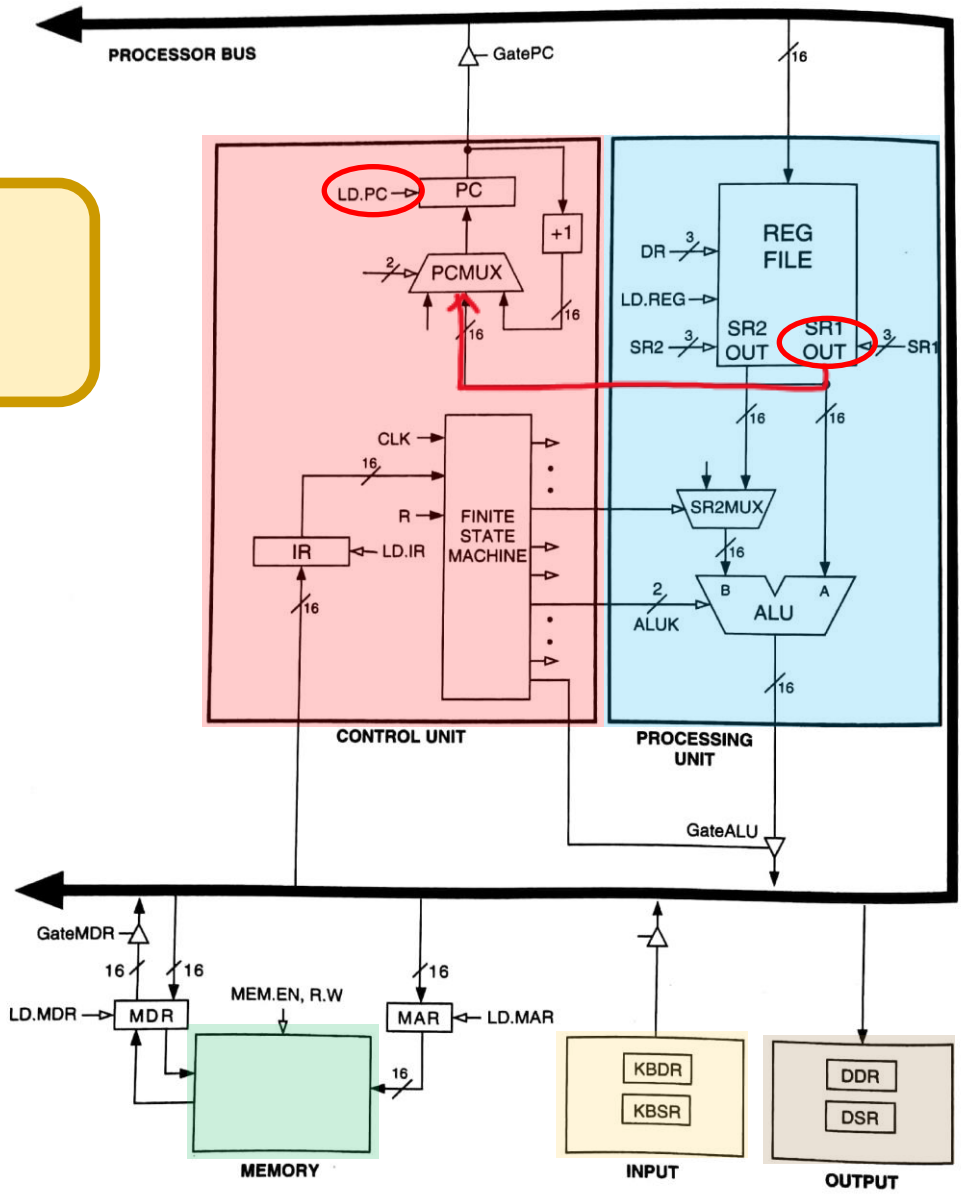
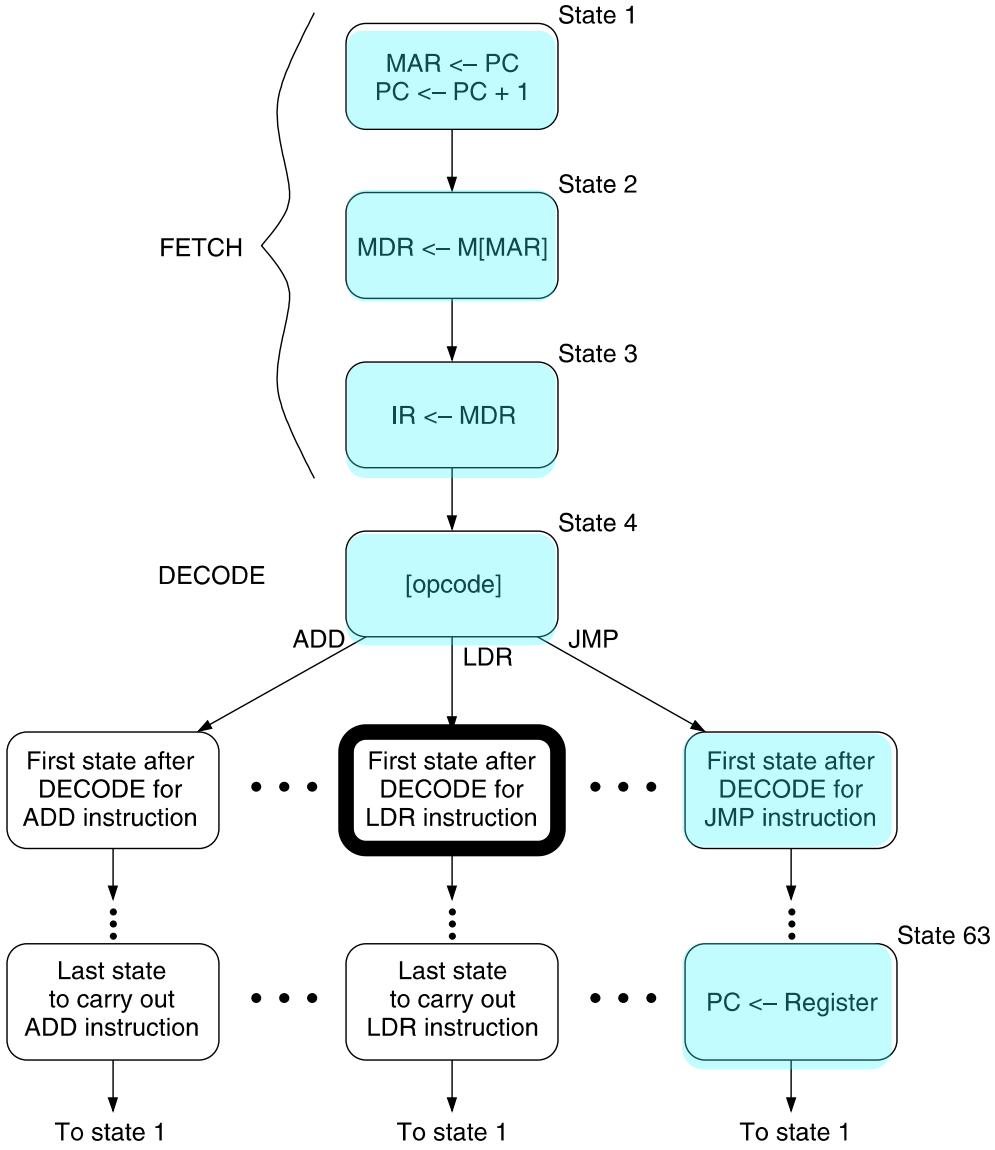


Figure 4.3 The LC-3 as an example of the von Neumann model

Control of the Instruction Cycle



- State 1
  - The FSM asserts GatePC and LD.MAR
  - It selects input (+1) in PCMUX and asserts LD.PC
- State 2
  - MDR is loaded with the instruction
- State 3
  - The FSM asserts GateMDR and LD.IR
- State 4
  - The FSM goes to next state depending on opcode
- State 63
  - JMP loads register into PC
- Full state diagram in Patt&Pattel, Appendix C

Figure 4.4 An abbreviated state diagram of the LC-3

This is an FSM Controlling the LC-3 Processor

To Come: Full State Machine for LC-3b

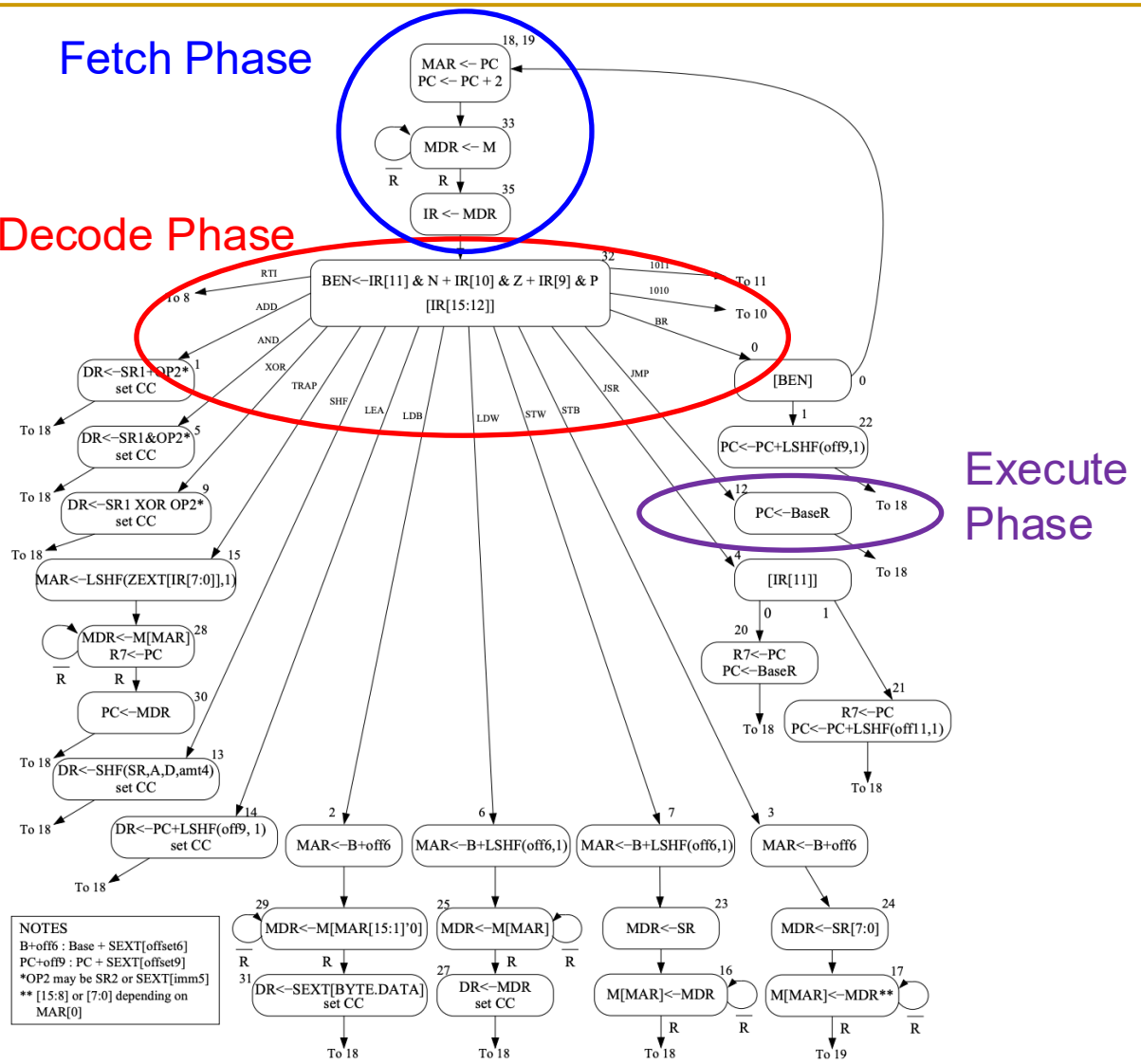


Figure C.2: A state machine for the LC-3b

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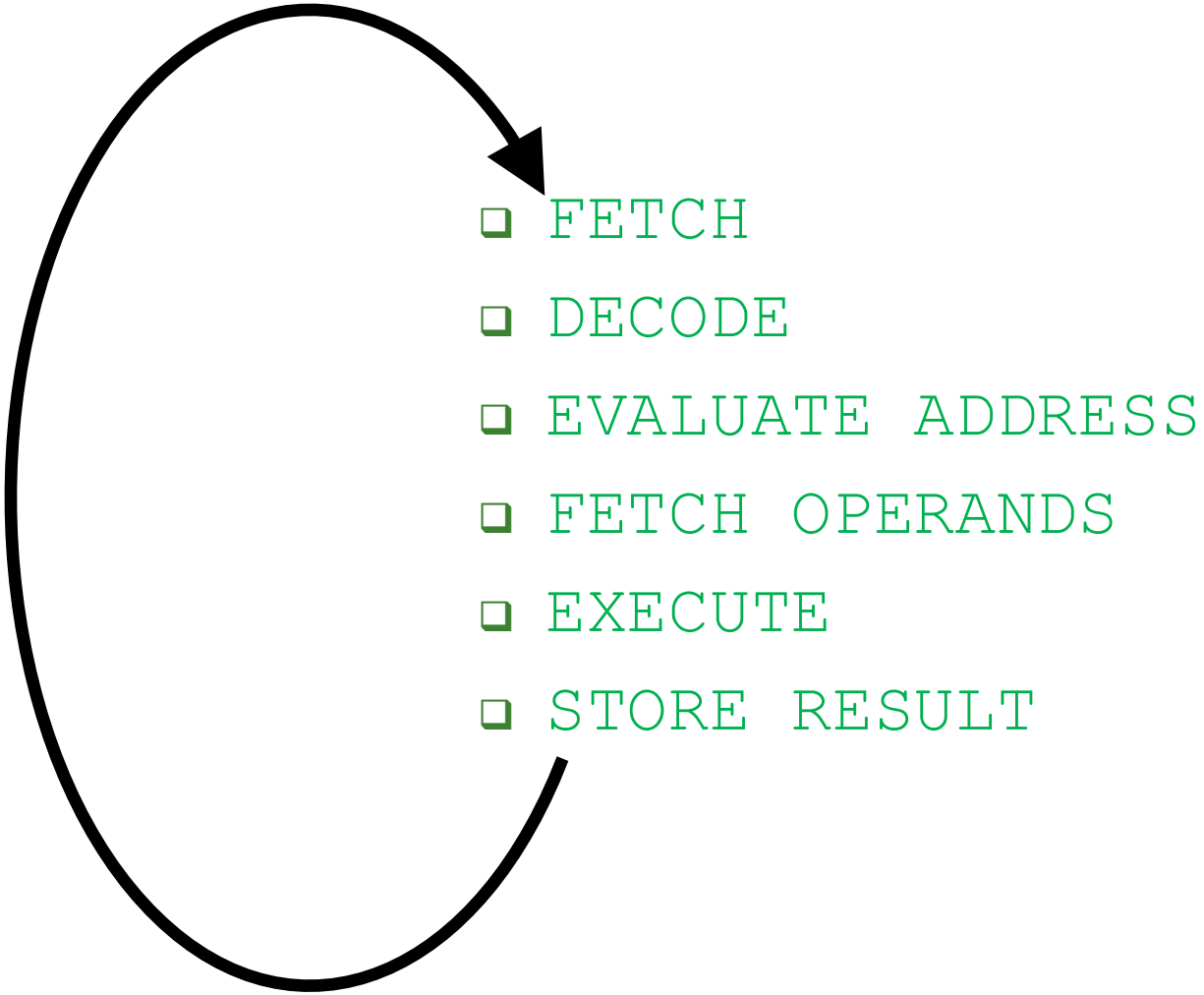
The Instruction Cycle



- FETCH
- DECODE
- EVALUATE ADDRESS
- FETCH OPERANDS
- EXECUTE
- STORE RESULT

LC-3 and MIPS Instruction Set Architectures

Agenda for Today & Next Few Lectures

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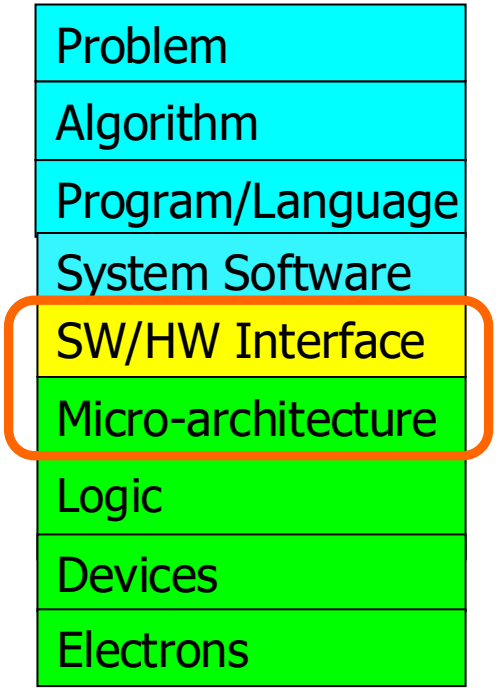
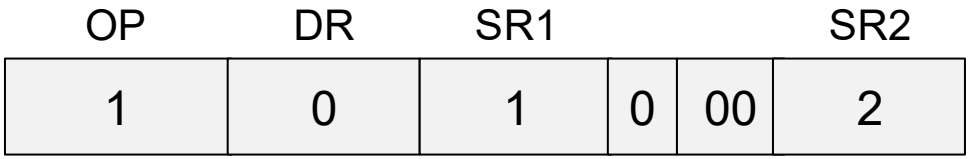


The Instruction Set

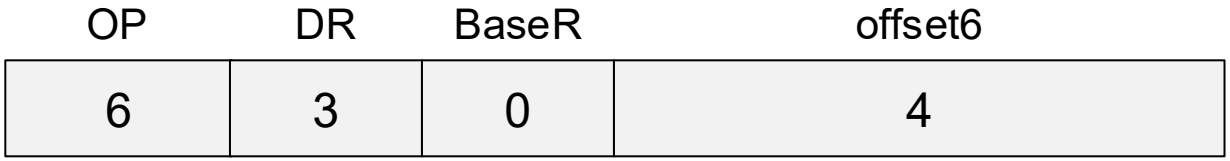
- It defines opcodes, data types, and addressing modes
- ADD and LDR have been our first examples

ADD



Register mode

LDR

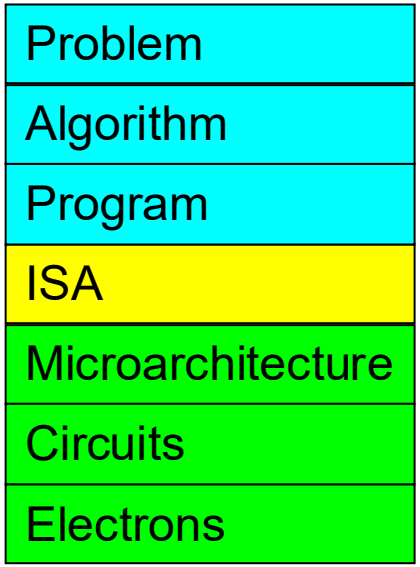


Base+offset mode

The Instruction Set Architecture

- The ISA is the interface between what the software commands and what the hardware carries out

- The ISA specifies
  - The memory organization
    - Address space (LC-3: 2^{16}, MIPS: 2^{32})
    - Addressability (LC-3: 16 bits, MIPS: 8 bits)
      - Word- or Byte-addressable
  - The register set
    - 8 registers (R0 to R7) in LC-3
    - 32 registers in MIPS
  - The instruction set
    - Opcodes
    - Data types
    - Addressing modes
    - Length and format of instructions



Instructions (Opcodes)

Opcodes

- A large or small set of opcodes could be defined
  - E.g, HP Precision Architecture: an instruction for A*B+C
  - E.g, x86 ISA: multimedia extensions (MMX), later SSE, AVX, AMX
  - E.g, VAX ISA: opcode to save all information of one program prior to switching to another program

- Tradeoffs are involved. Examples:
  - Hardware complexity vs. software complexity
  - Latency of simple vs. complex instructions

- In LC-3 and in MIPS there are three types of opcodes
  - Operate
  - Data movement
  - Control

Opcodes in LC-3

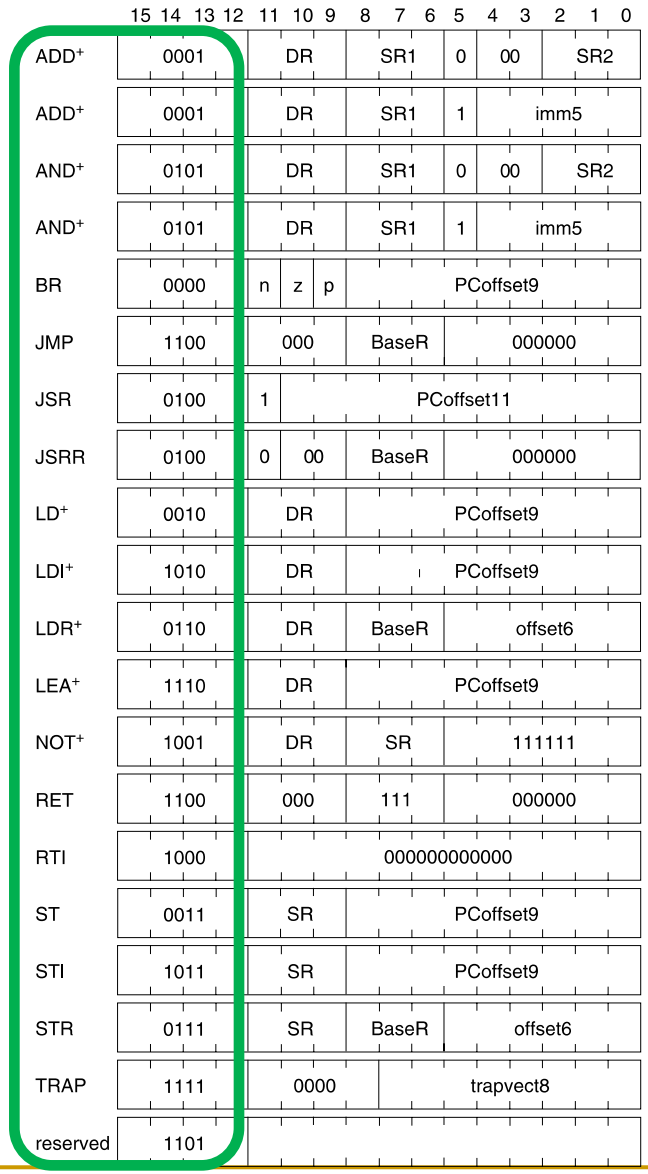
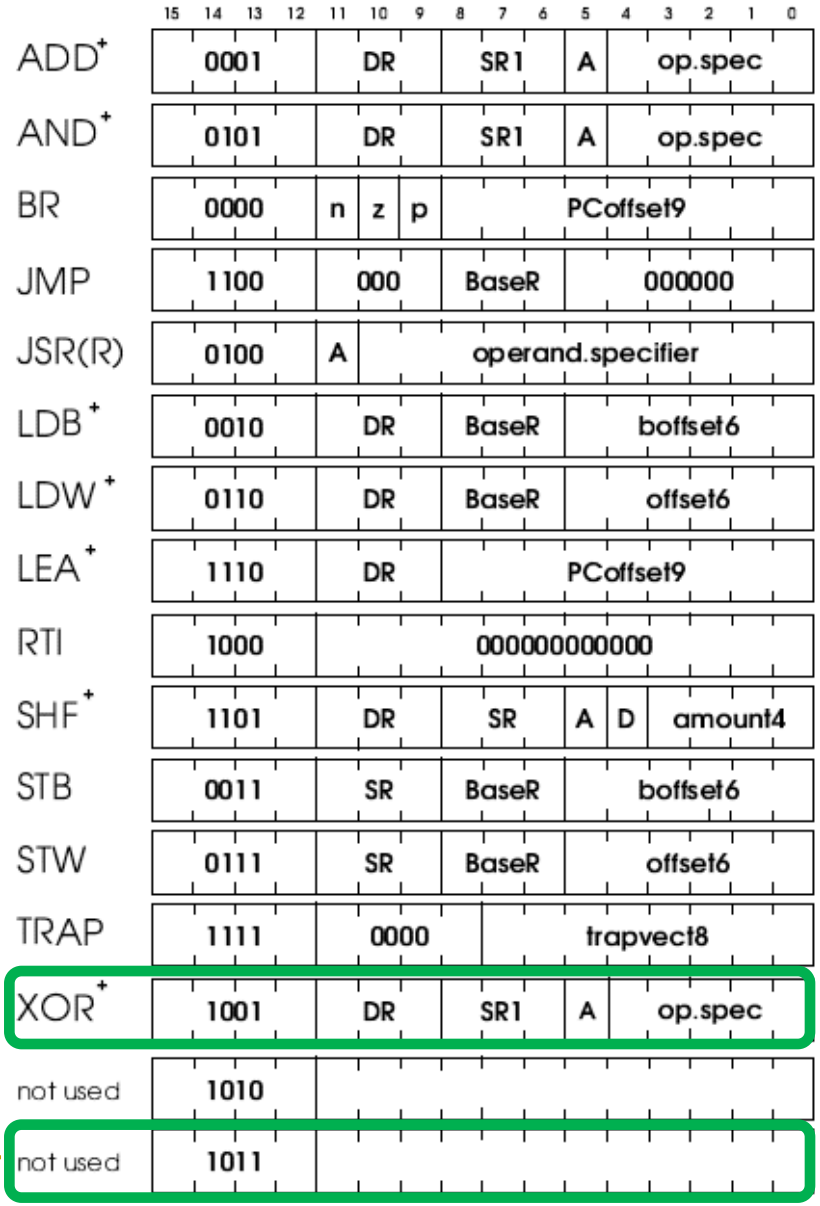
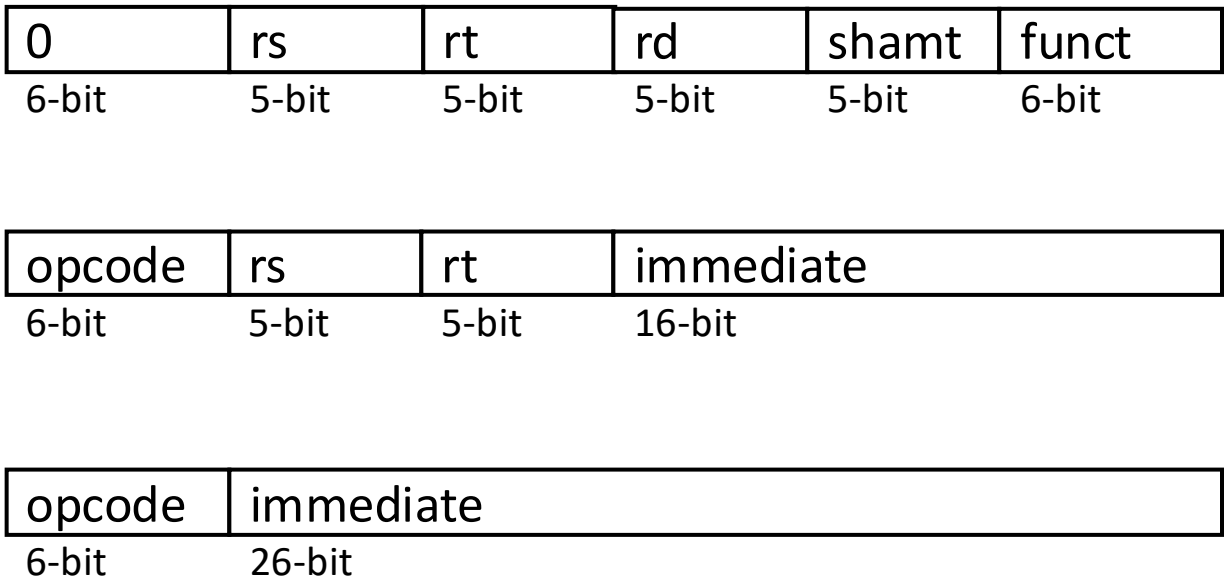


Figure 5.3 Formats of the entire LC-3 instruction set. NOTE: + indicates instructions that modify condition codes

Opcodes in LC-3b



MIPS Instruction Types



Funct in MIPS R-Type Instructions (I)

Opcode is 0 in MIPS R-Type instructions.

Funct defines the operation

Table B.2 R-type instructions, sorted by funct field

Funct	Name	Description	Operation
000000 (0)	`sll rd, rt, shamt`	shift left logical	`[rd] = [rt] << shamt`
000010 (2)	`srl rd, rt, shamt`	shift right logical	`[rd] = [rt] >> shamt`
000011 (3)	`sra rd, rt, shamt`	shift right arithmetic	`[rd] = [rt] >>> shamt`
000100 (4)	`sllv rd, rt, rs`	shift left logical variable	`[rd] = [rt] << [rs]`$_{4:0}$
000110 (6)	`srlv rd, rt, rs`	shift right logical variable	`[rd] = [rt] >> [rs]`$_{4:0}$
000111 (7)	`srav rd, rt, rs`	shift right arithmetic variable	`[rd] = [rt] >>> [rs]`$_{4:0}$
001000 (8)	`jr rs`	jump register	`PC = [rs]`
001001 (9)	`jalr rs`	jump and link register	`$ra = PC + 4, PC = [rs]`
001100 (12)	`syscall`	system call	`system call exception`
001101 (13)	`break`	break	`break exception`
010000 (16)	`mfhi rd`	move from hi	`[rd] = [hi]`
010001 (17)	`mthi rs`	move to hi	`[hi] = [rs]`
010010 (18)	`mflo rd`	move from lo	`[rd] = [lo]`
010011 (19)	`mtlo rs`	move to lo	`[lo] = [rs]`
011000 (24)	`mult rs, rt`	multiply	`{[hi], [lo]} = [rs] × [rt]`
011001 (25)	`multu rs, rt`	multiply unsigned	`{[hi], [lo]} = [rs] × [rt]`
011010 (26)	`div rs, rt`	divide	`[lo] = [rs]/[rt], [hi] = [rs]%[rt]`
011011 (27)	`divu rs, rt`	divide unsigned	`[lo] = [rs]/[rt], [hi] = [rs]%[rt]`

(*continued*)

Funct in MIPS R-Type Instructions (II)

Table B.2 R-type instructions, sorted by funct field—Cont'd

Funct	Name	Description	Operation
100000 (32)	`add rd, rs, rt`	add	`[rd] = [rs] + [rt]`
100001 (33)	`addu rd, rs, rt`	add unsigned	`[rd] = [rs] + [rt]`
100010 (34)	`sub rd, rs, rt`	subtract	`[rd] = [rs] - [rt]`
100011 (35)	`subu rd, rs, rt`	subtract unsigned	`[rd] = [rs] - [rt]`
100100 (36)	`and rd, rs, rt`	and	`[rd] = [rs] & [rt]`
100101 (37)	`or rd, rs, rt`	or	`[rd] = [rs] | [rt]`
100110 (38)	`xor rd, rs, rt`	xor	`[rd] = [rs] ^ [rt]`
100111 (39)	`nor rd, rs, rt`	nor	`[rd] = ~([rs] | [rt])`
101010 (42)	`slt rd, rs, rt`	set less than	`[rs] < [rt] ? [rd] = 1 : [rd] = 0`
101011 (43)	`sltu rd, rs, rt`	set less than unsigned	`[rs] < [rt] ? [rd] = 1 : [rd] = 0`

- More complete list of instructions are in H&H Appendix B

Data Types

Data Types

- An ISA supports one or several data types

- LC-3 only supports 2's complement integers **H&H Chapter 1.4.6**
  - Negative of a 2's complement binary value X = NOT(X) + 1

- MIPS supports
  - 2's complement integers
  - Unsigned integers
  - Floating point

- Tradeoffs are involved. Examples:
  - Hardware complexity vs. software complexity
  - Latency of operations on supported vs. unsupported data types

Why Have Different Data Types in ISA?

- An example of programmer vs. microarchitect tradeoff

- Advantage of more data types:
  - Enables better mapping of high-level programming constructs to hardware
    - Hardware can directly operate on data types present in programming languages → small number of instructions and code size
      - Matrix operations vs. individual multiply/add/load/store instructions
      - Graph operations vs. individual load/store/add/… instructions

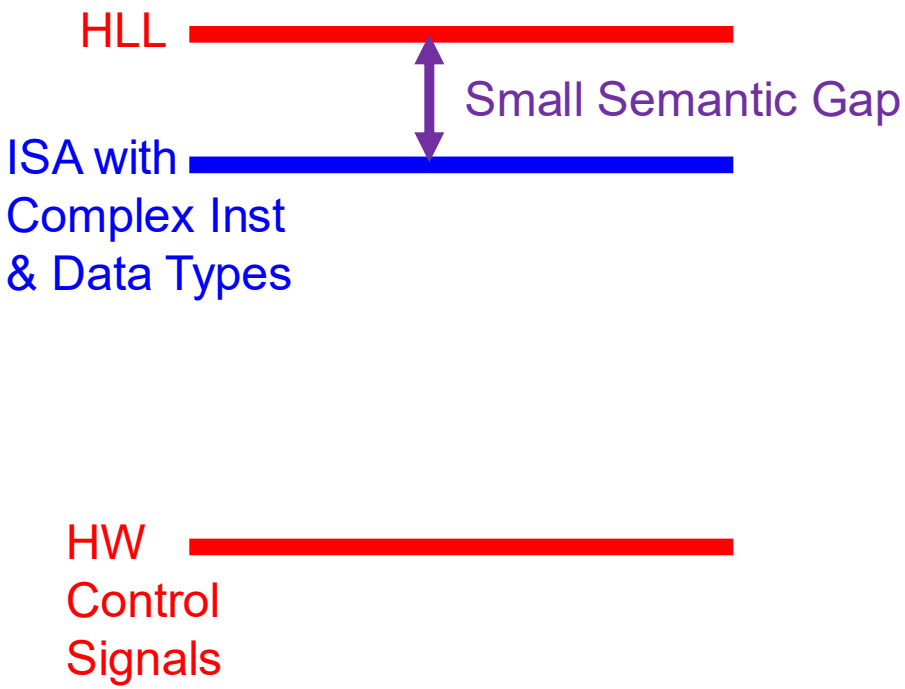
- Disadvantage:
  - More work for the microarchitect
    - who needs to implement the data types and instructions that operate on data types

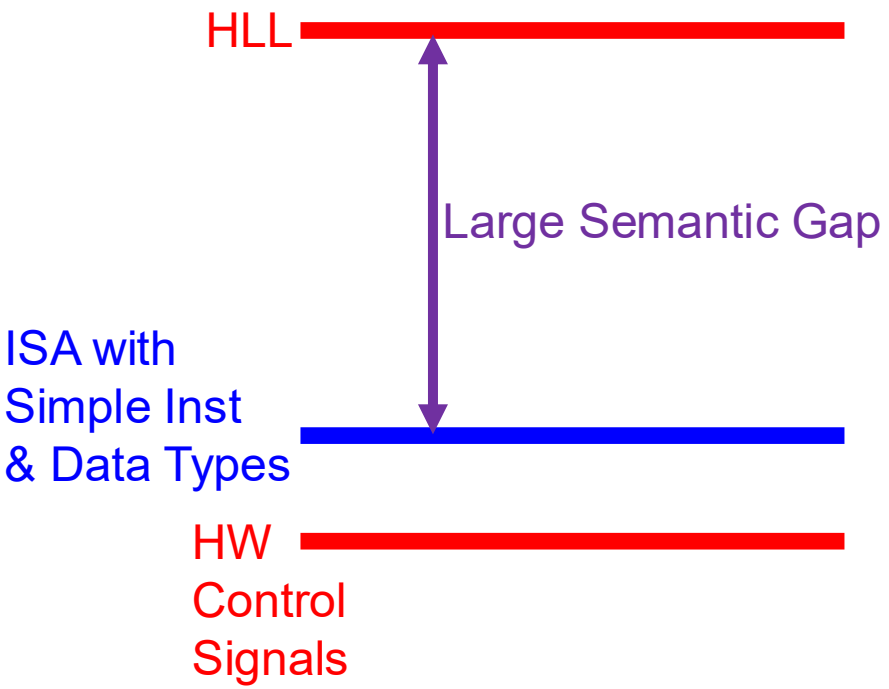
Data Types and Instruction Complexity

- Data types are coupled tightly to the semantic level, or complexity of instructions
- Concept of semantic gap
  - how close instructions & data types are to high-level language
- Complex instructions + data types → small semantic gap
  - E.g., insert into a doubly linked list, multiply two matrices
  - VAX ISA: doubly-linked list, multi-dimensional arrays
- Simple instructions + data types → large semantic gap
  - E.g., primitive operations: load, store, multiply, add, nor
  - Early RISC machines: Only integer data type, simple operations

Semantic Gap

- How close instructions & data types are to high-level language (HLL)





Complex vs. Simple Instructions+Data Types

- Complex instruction: An instruction does a lot of work, e.g. many operations
  - Insert in a doubly linked list
  - Compute FFT
  - String copy
  - Matrix multiply
  - …
- Simple instruction: An instruction does little work -- it is a primitive using which complex operations can be built
  - Add
  - XOR
  - Multiply
  - …

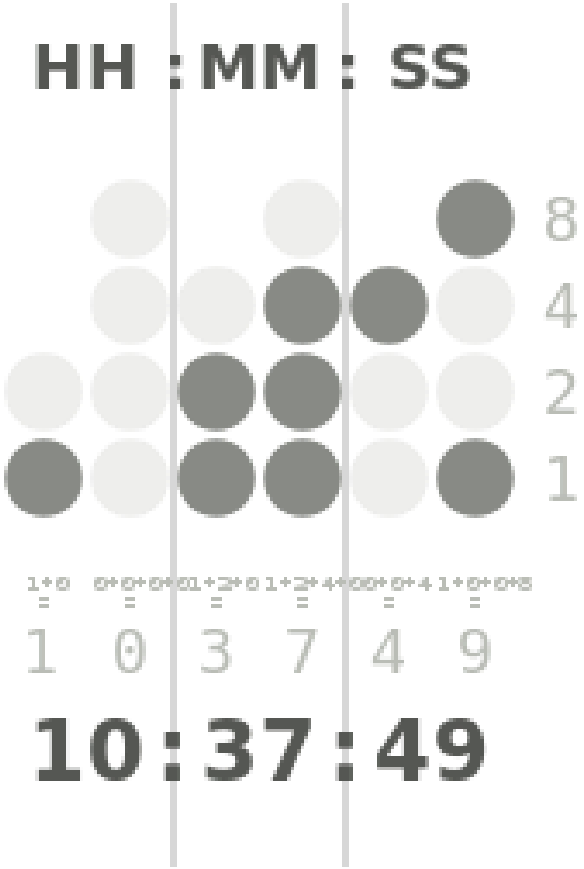
Complex vs. Simple Instructions+Data Types

- Advantages of Complex Instructions + Data Types
  - + Denser encoding → smaller code size → better memory utilization, saves off-chip bandwidth, better cache hit rate (better packing of instructions)
  - + Simpler compiler: no need to optimize small instructions as much

- Disadvantages of Complex Instructions + Data Types
  - - Larger chunks of work → compiler has less opportunity to optimize (limited in fine-grained optimizations it can do)
  - - More complex hardware → translation from a high level to control signals and optimization needs to be done by hardware

Aside: An Example: **B**inary**C**oded**D**ecimal

- Each decimal digit is encoded with a fixed number of bits



"Binary clock" by Alexander Jones & Eric Pierce - Own work, based on Wapcaplet's Binary clock.png on the English Wikipedia. Licensed under CC BY-SA 3.0 via Wikimedia Commons - http://commons.wikimedia.org/wiki/File:Binary_clock.svg#mediaviewer/File:Binary_clock.svg

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Aside: An Example: **B**inary**C**oded**D**ecimal

- Each decimal digit is encoded with a fixed number of bits

Addressing Modes

Addressing Modes

- An addressing mode is a mechanism for specifying where an operand is located

- There are five addressing modes in LC-3
  - Immediate or literal (constant)
    - The operand is in some bits of the instruction
  - Register
    - The operand is in one of R0 to R7 registers
  - Three memory addressing modes
    - PC-relative
    - Indirect
    - Base+offset

- MIPS has pseudo-direct addressing (for j and jal), additionally, but does **not** have indirect addressing

Why Have Different Addressing Modes?

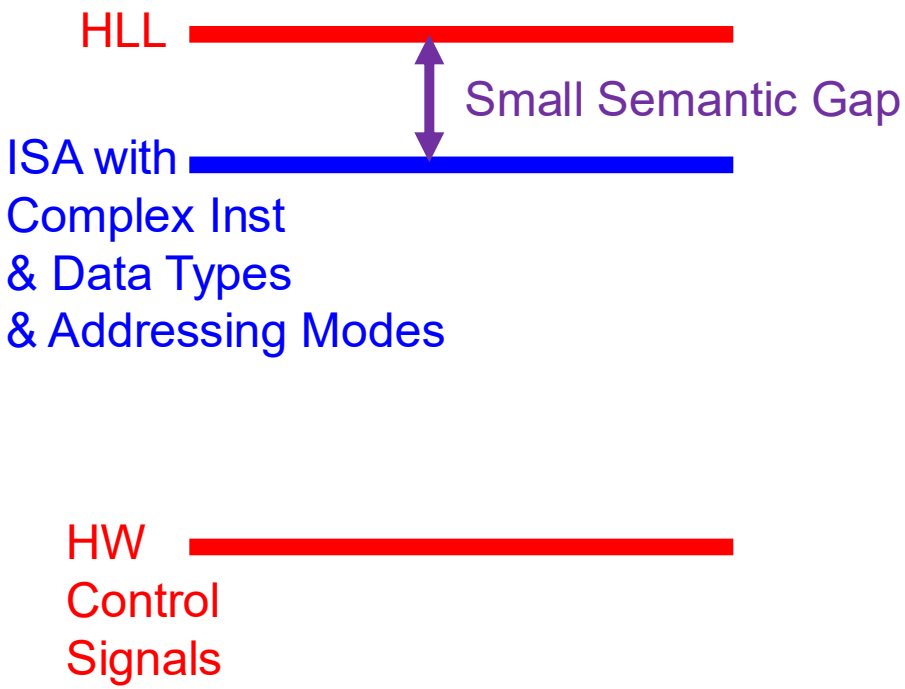
- Another example of programmer vs. microarchitect tradeoff

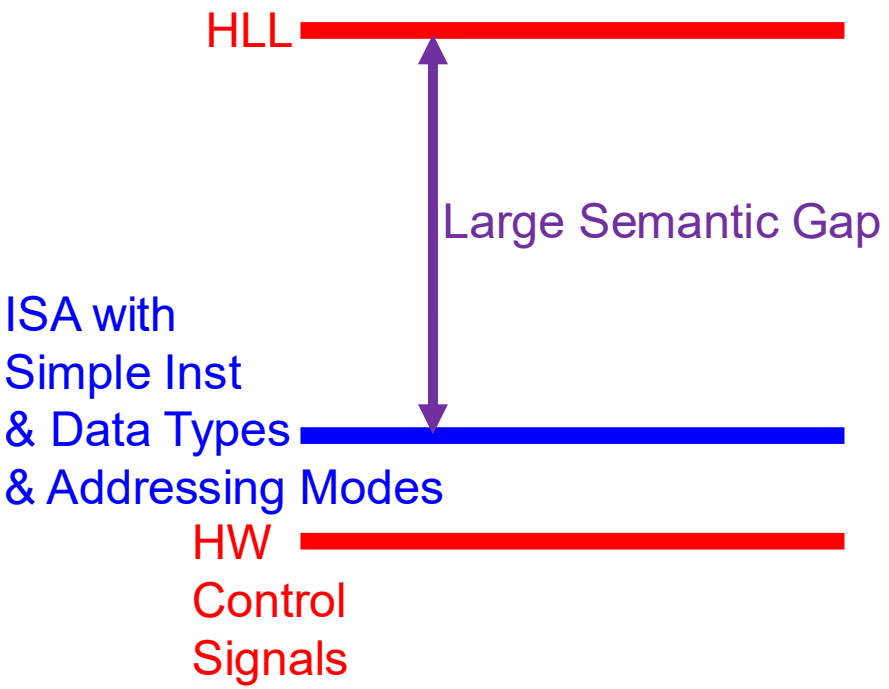
- Advantage of more addressing modes:
  - Enables better mapping of high-level programming constructs to hardware
    - some accesses are better expressed with a different mode → reduced number of instructions and code size
      - Array indexing (one or multi-dimensional)
      - Pointer-based accesses (indirection)
      - Matrix element indexing; sparse matrix element indexing

- Disadvantages:
  - More work for the microarchitect
  - More options for the compiler to decide what to use

Semantic Gap Applies to Addressing Modes

- How close instructions & data types & addressing modes are to high-level language (HLL)





Many Tradeoffs in ISA Design...

- Execution model – sequencing model and processing style
- Instruction length
- Instruction format
- Instruction types
- Instruction complexity vs. simplicity
- Data types
- Number of registers
- Addressing mode types
- Memory organization (address space, addressability, endianness, …)
- Memory access restrictions and permissions
- Support for multiple instructions to execute in parallel?
- …

Operate Instructions

Operate Instructions

- In LC-3, there are three operate instructions
  - NOT is a unary operation (one source operand)
    - It executes bitwise NOT
  - ADD and AND are binary operations (two source operands)
    - ADD is 2's complement addition
    - AND is bitwise SR1 & SR2

- In MIPS, there are many more
  - Many R-type instructions (they are binary operations)
    - E.g., add, and, nor, xor…
  - I-type versions (i.e., with one immediate operand) of the R-type operate instructions
  - F-type operations, i.e., floating-point operations

NOT in LC-3

- NOT assembly and machine code

LC-3 assembly

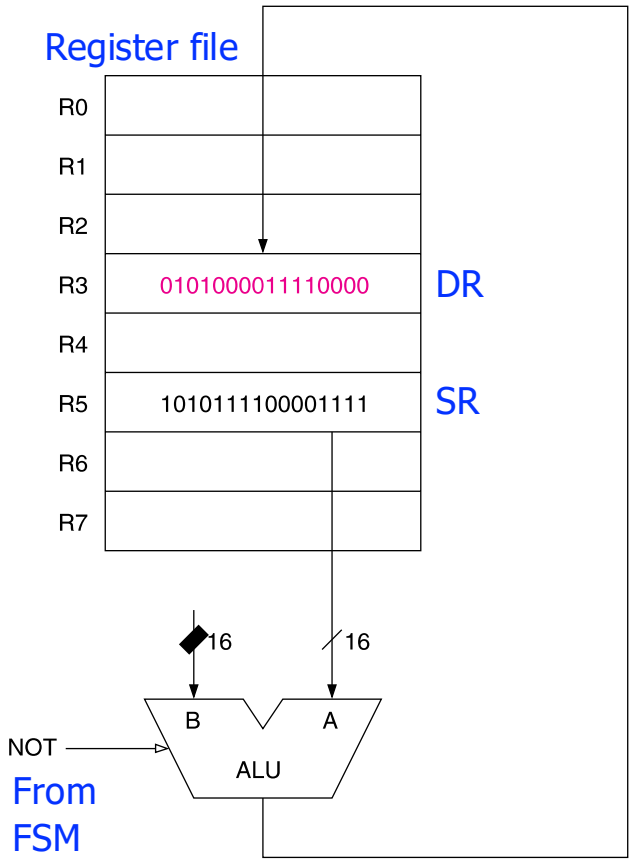
```
NOT   R3, R5
```

Field Values

OP	DR	SR	
9	3	5	1 1 1 1 1 1

Machine Code

OP	DR	SR	
1 0 0 1	0 1 1	0 0 1	1 1 1 1 1 1
15 12	11 9	8 6	5 0



There is no NOT in MIPS. How is it implemented?

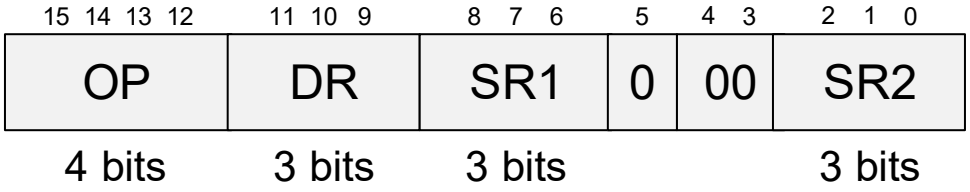
There is no NOT in LC-3b. How is it implemented?

Operate Instructions

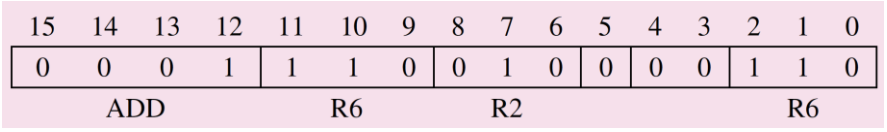
- We are already familiar with LC-3's ADD and AND with register mode (R-type in MIPS)
- Now let us see the versions with one literal (i.e., immediate) operand
- We will use Subtraction as an example
  - How is it implemented in LC-3 and MIPS?

Recall: LC-3 Operate Instruction Format

- LC-3 Operate Instruction Format (Register OP Register)

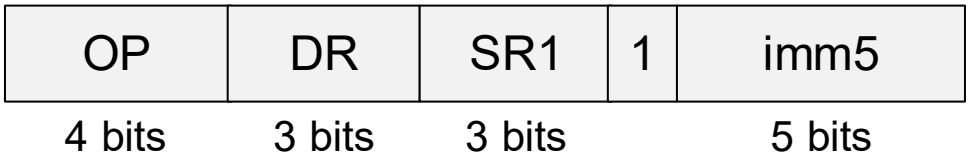


  - OP = opcode (what the instruction does)
    - E.g., ADD = 0001
      - Semantics: DR ← SR1 + SR2
    - E.g., AND = 0101
      - Semantics: DR ← SR1 AND SR2
  - SR1, SR2 = source registers
  - DR = destination register



Operate Instr. with one Literal in LC-3

- ADD and AND



  - OP = operation
    - E.g., ADD = 0001 (same OP as the register-mode ADD)
      - DR ← SR1 + sign-extend(imm5)
    - E.g., AND = 0101 (same OP as the register-mode AND)
      - DR ← SR1 AND sign-extend(imm5)
  - SR1 = source register
  - DR = destination register
  - imm5 = Literal or immediate (sign-extend to 16 bits)

ADD with one Literal in LC-3

- ADD assembly and machine code

LC-3 assembly

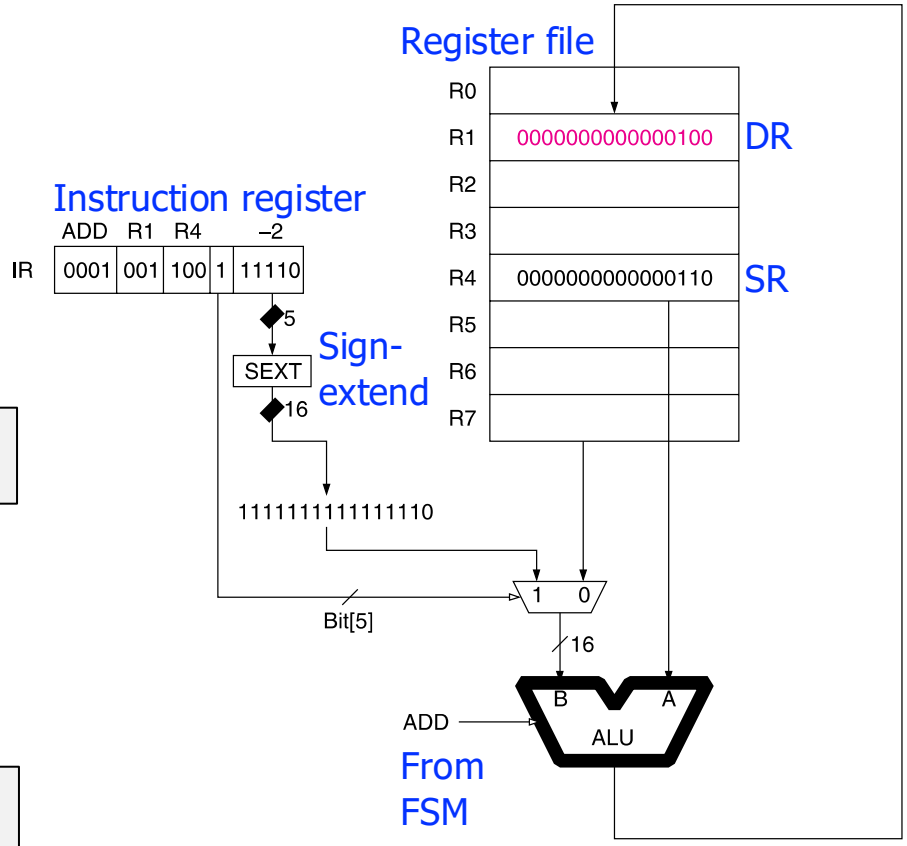
```
ADD R1, R4, #-2
```

Field Values

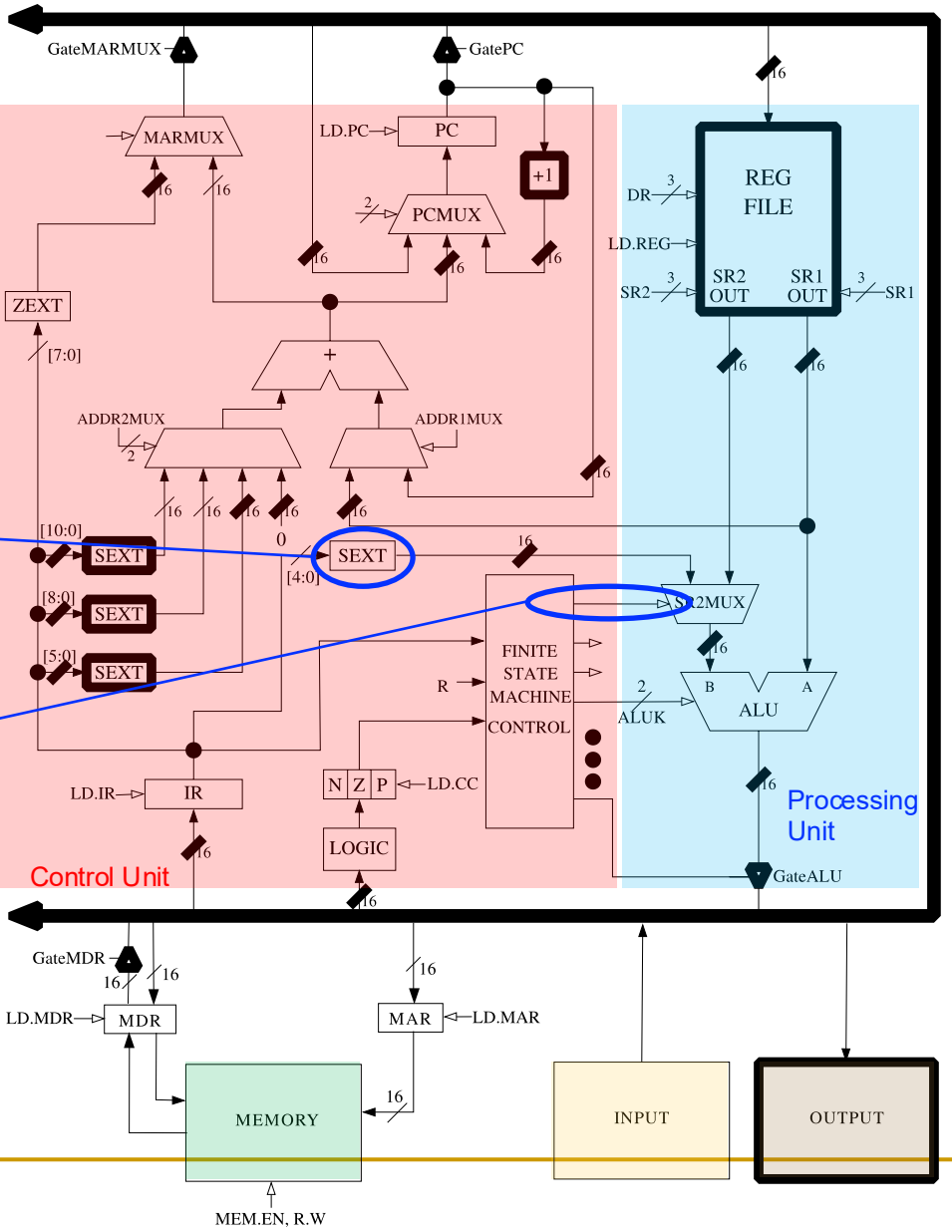
OP	DR	SR		imm5
1	1	4	1	-2

Machine Code

OP	DR	SR		imm5
0 0 0 1	0 0 1	1 0 0	1	1 1 1 1 0
15 12	11 9	8 6	5	4 0



ADD with one Literal in LC-3 Data Path

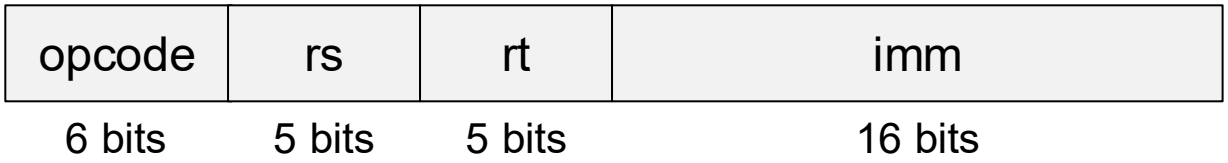


Sign extension (Operand)

Select Immediate or Register (as the 2nd input to instruction)

Instructions with one Literal in MIPS

- I-type MIPS Instructions
  - 2 register operands and immediate
- Some operate and data movement instructions



  - opcode = operation
  - rs = source register
  - rt =
    - destination register in some instructions (e.g., `addi`, `lw`)
    - source register in others (e.g., `sw`)
  - imm = Literal or immediate

ADD with one Literal in MIPS

- Add immediate

MIPS assembly

```
addi $s0, $s1, 5
```

Field Values

op	rs	rt	imm
8	17	16	5

rt ← rs + sign-extend(imm)

Machine Code

op	rs	rt	imm
001000	10001	10010	0000 0000 0000 0101

0x22300005

Subtraction in MIPS vs. LC-3

- MIPS assembly

High-level code

```
a = b + c - d;
```

MIPS assembly

```
add  $t0, $s0, $s1
sub  $s3, $t0, $s2
```

- LC-3 assembly

High-level code

```
a = b + c - d;
```

LC-3 assembly

```
ADD  R2, R0, R1
NOT  R4, R3
ADD  R5, R4, #1
ADD  R6, R2, R5
```

2's complement of R3

- Tradeoff in LC-3
  - More instructions
  - But, simpler control logic

Subtract Immediate

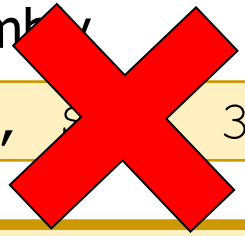
- MIPS assembly

High-level code

```
a = b - 3;
```

MIPS assembly

```
subi $s1, $s0, 3
```



Is subi necessary in MIPS?

MIPS assembly

```
addi $s1, $s0, -3
```

- LC-3

High-level code

```
a = b - 3;
```

LC-3 assembly

```
ADD  R1, R0, #-3
```

Data Movement Instructions and Addressing Modes

Data Movement Instructions

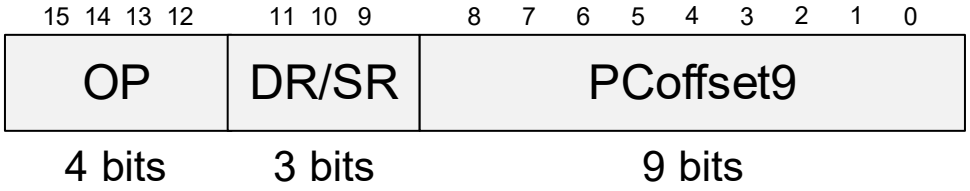
- In LC-3, there are seven data movement instructions
  - LD, LDR, LDI, LEA, ST, STR, STI

- Format of load and store instructions
  - Opcode (bits [15:12])
  - DR or SR (bits [11:9])
  - Address generation bits (bits [8:0])
  - Four ways to interpret bits, called addressing modes
    - PC-Relative Mode
    - Indirect Mode
    - Base+Offset Mode
    - Immediate Mode

- In MIPS, there are only Base+offset and Immediate modes for load and store instructions

PC-Relative Addressing Mode

- LD (Load) and ST (Store)



  - OP = opcode
    - E.g., LD = 0010
    - E.g., ST = 0011
  - DR = destination register in LD
  - SR = source register in ST
  - LD: DR ← Memory[PC† + sign-extend(PCoffset9)]
  - ST: Memory[PC† + sign-extend(PCoffset9)] ← SR

† This is the incremented PC

LD in LC-3

- LD assembly and machine code

LC-3 assembly

```
LD R2, 0x1AF
```

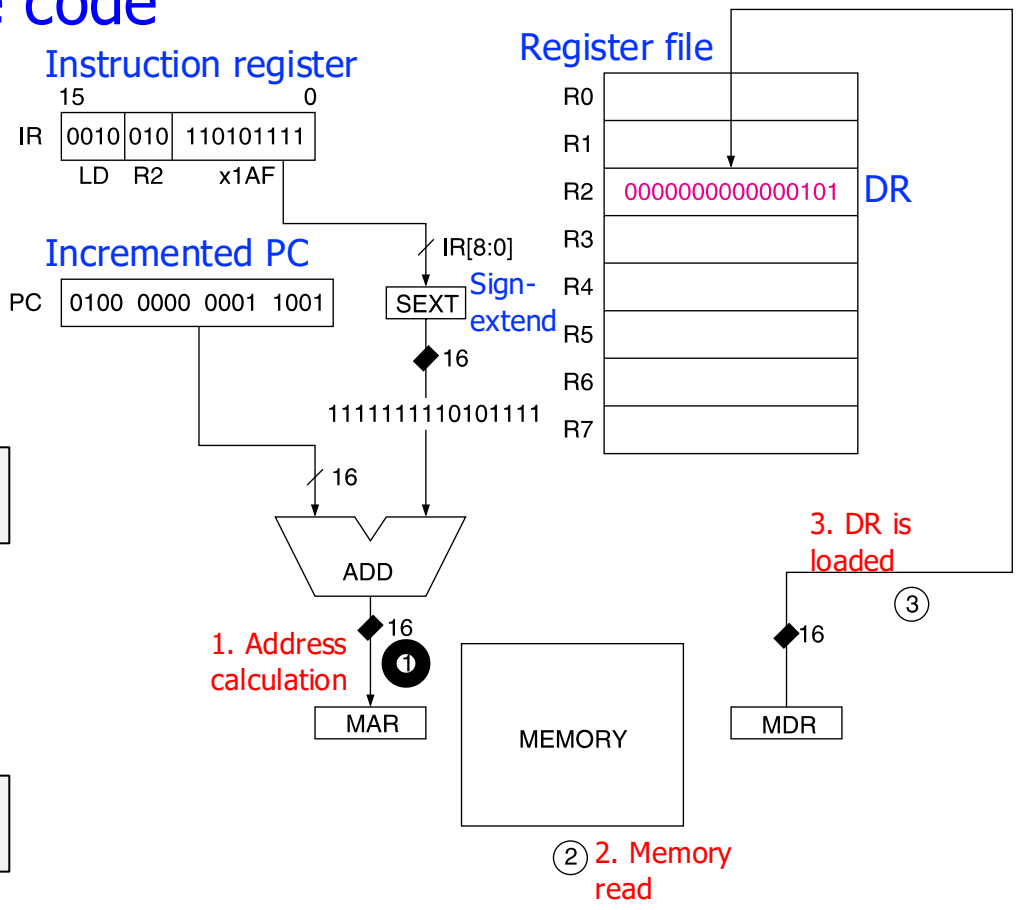
Field Values

OP	DR	PCoffset9
2	2	0x1AF

Machine Code

OP	DR	PCoffset9
0 0 1 0	0 1 0	1 1 0 1 0 1 1 1 1

15 12 11 9 8 0

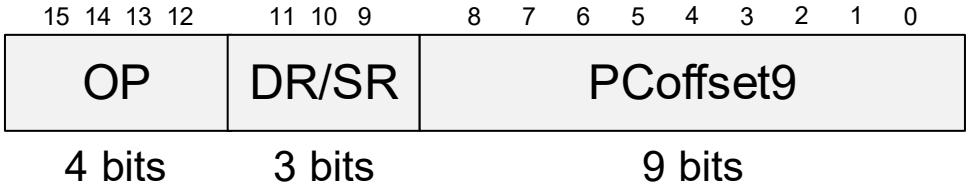


The memory address is only +255 to -256 locations away of the LD or ST instruction

Limitation: The PC-relative addressing mode cannot address far away from the instruction

Indirect Addressing Mode

- LDI (Load Indirect) and STI (Store Indirect)



  - OP = opcode
    - E.g., LDI = 1010
    - E.g., STI = 1011
  - DR = destination register in LDI
  - SR = source register in STI
  - LDI: DR ← Memory[Memory[PC† + sign-extend(PCoffset9)]]
  - STI: Memory[Memory[PC† + sign-extend(PCoffset9)]] ← SR

† This is the incremented PC

LDI in LC-3

- LDI assembly and machine code

LC-3 assembly

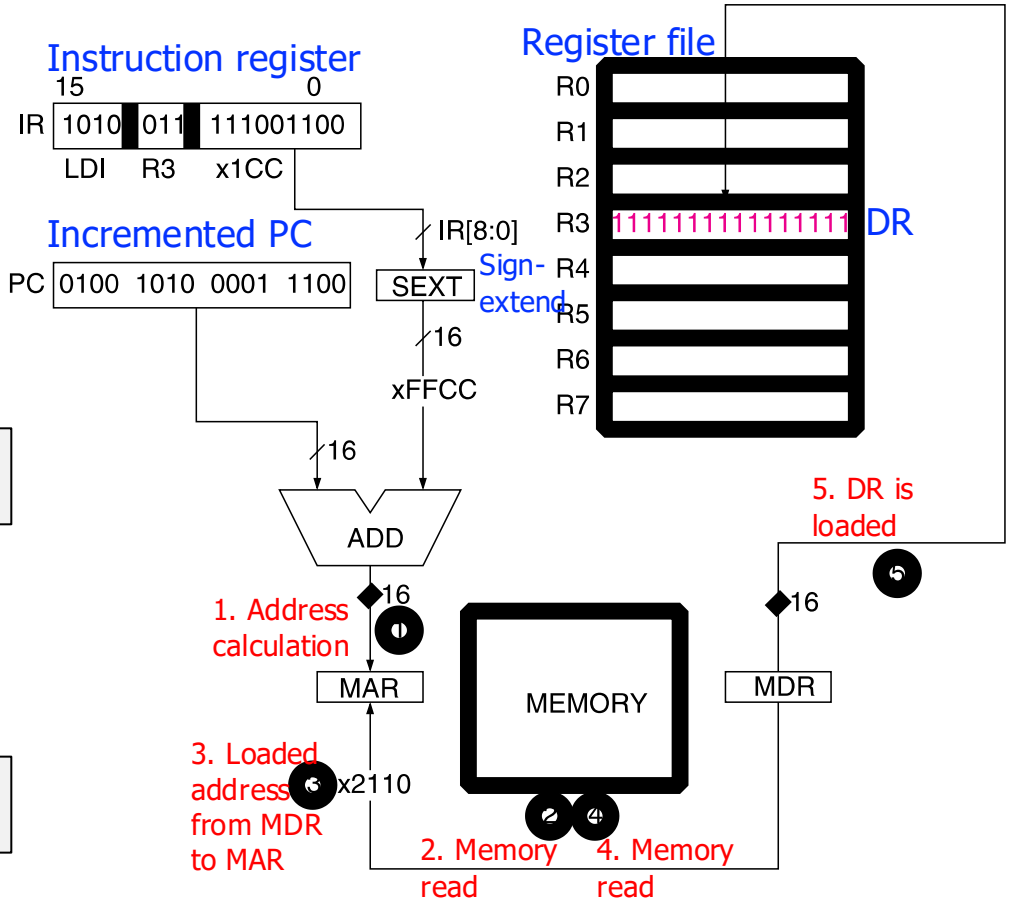
```
LDI R3, 0x1CC
```

Field Values

OP	DR	PCoffset9
A	3	0x1CC

Machine Code

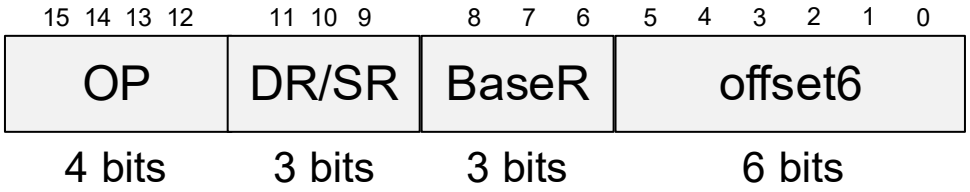
OP	DR	PCoffset9
1 0 1 0	0 1 1	1 1 1 0 0 1 1 0 0
15 12	11 9	8 0



Now the address of the operand can be anywhere in the memory

Base+Offset Addressing Mode

- LDR (Load Register) and STR (Store Register)



  - OP = opcode
    - E.g., LDR = 0110
    - E.g., STR = 0111
  - DR = destination register in LDR
  - SR = source register in STR
  - LDR: DR ← Memory[BaseR + sign-extend(offset6)]
  - STR: Memory[BaseR + sign-extend(offset6)] ← SR

LDR in LC-3

- LDR assembly and machine code

LC-3 assembly

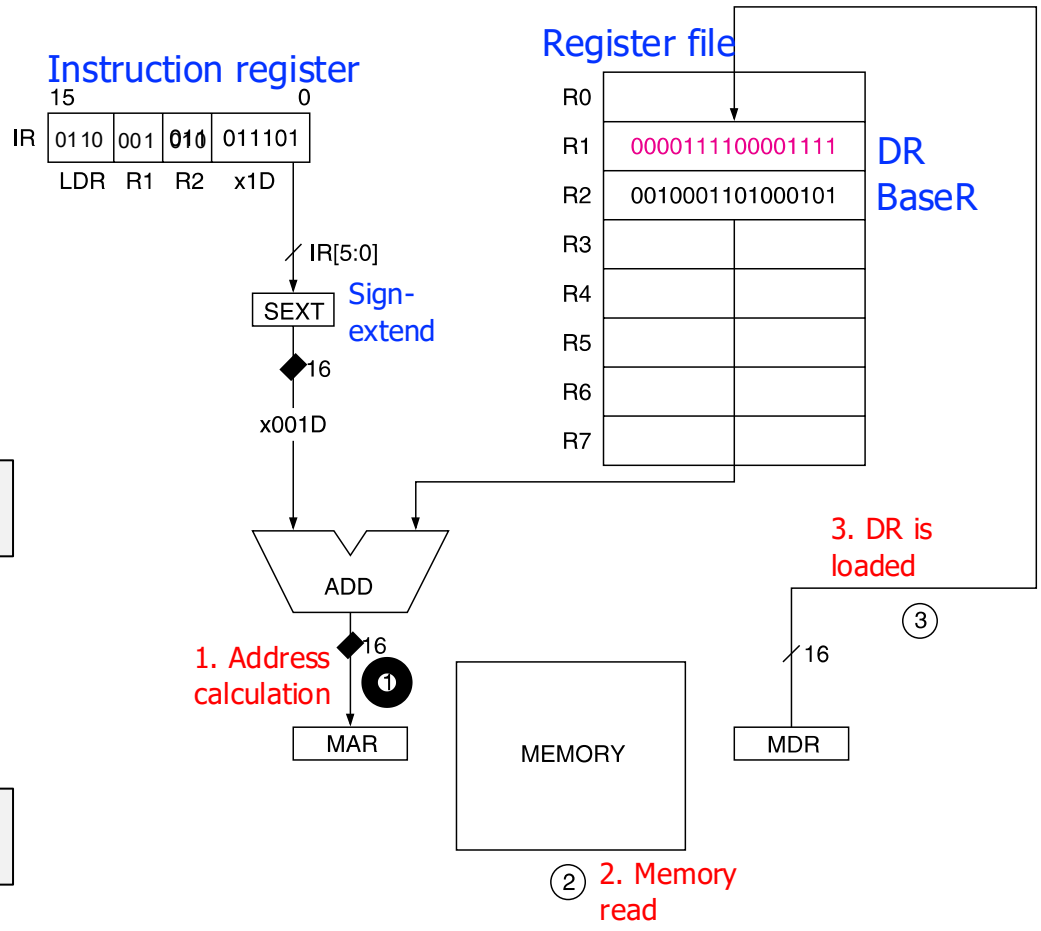
```
LDR R1, R2, 0x1D
```

Field Values

OP	DR	BaseR	offset6
6	1	2	0x1D

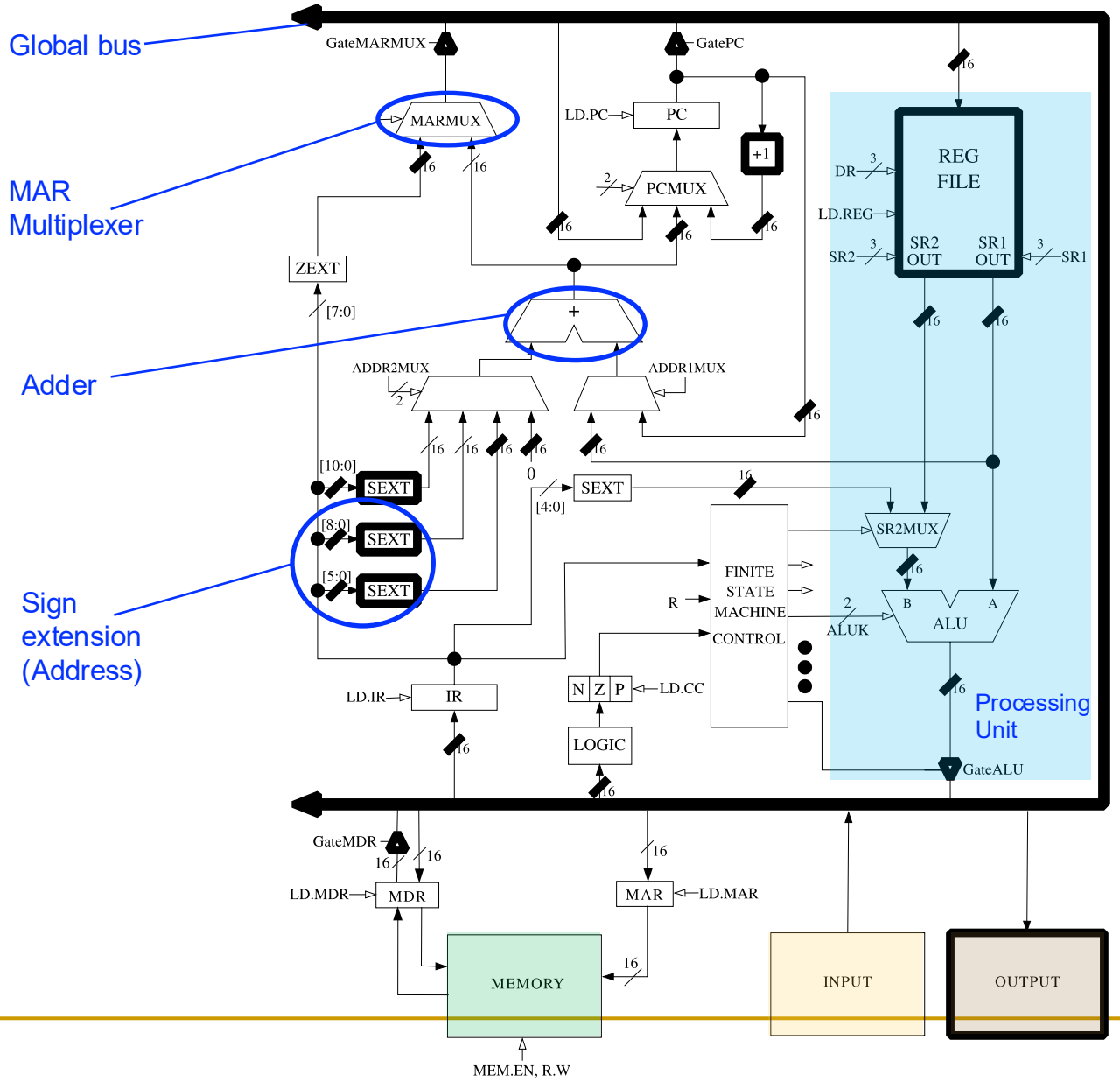
Machine Code

OP	DR	BaseR	offset6
0 1 1 0	0 0 1	0 1 0	0 1 1 1 0 1
15 12	11 9	8 6	5 0



Again, the address of the operand can be anywhere in the memory

Address Calculation in LC-3 Data Path



Base+Offset Addressing Mode in MIPS

- In MIPS, lw and sw use base+offset mode (or base addressing mode)

High-level code

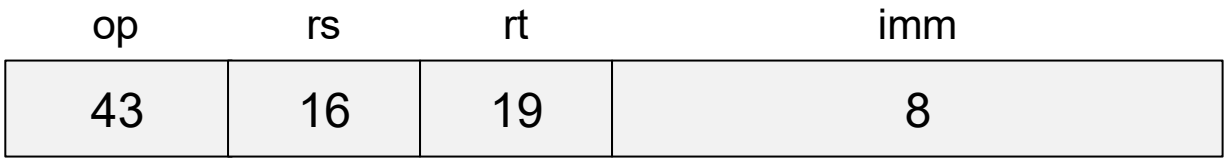
```
A[2] = a;
```

MIPS assembly

```
sw   $s3, 8($s0)
```

Memory[$s0 + 8] ← $s3

Field Values



- imm is the 16-bit offset, which is sign-extended to 32 bits

An Example Program in MIPS and LC-3

High-level code

```
a    = A[0];
c    = a + b - 5;
B[0] = c;
```

MIPS registers

```
A = $s0
b = $s2
B = $s1
```

LC-3 registers

```
A = R0
b = R2
B = R1
```

MIPS assembly

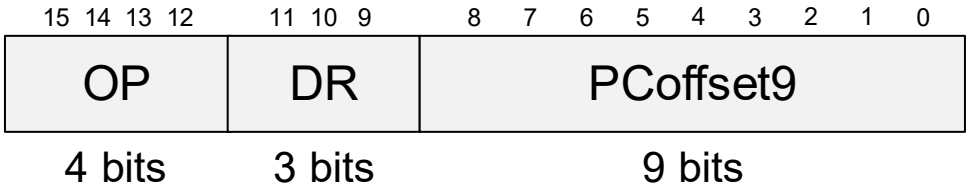
```
lw   $t0, 0($s0)
add  $t1, $t0, $s2
addi $t2, $t1, -5
sw   $t2, 0($s1)
```

LC-3 assembly

```
LDR  R5, R0, #0
ADD  R6, R5, R2
ADD  R7, R6, #-5
STR  R7, R1, #0
```

Immediate Addressing Mode (in LC-3)

- LEA (Load Effective Address)



  - OP = 1110
  - DR = destination register
  - LEA: DR ← PC† + sign-extend(PCoffset9)

What is the difference from PC-Relative addressing mode?

Answer: Instructions with PC-Relative mode load from memory, but LEA does not → Hence the name *Load Effective Address*

† This is the incremented PC

LEA in LC-3

- LEA assembly and machine code

LC-3 assembly

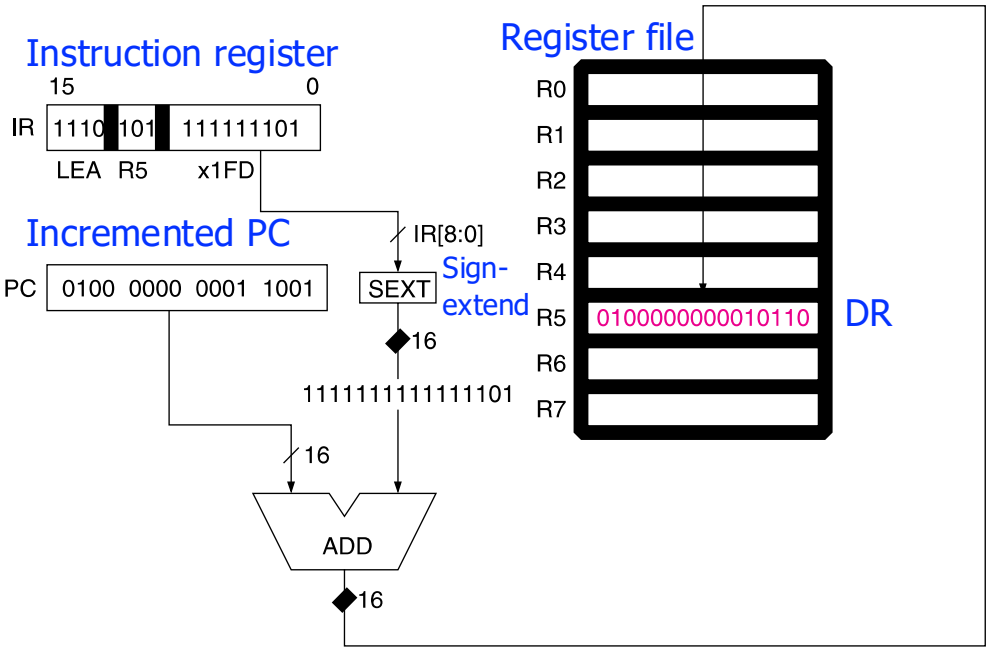
```
LEA R5, #-3
```

Field Values

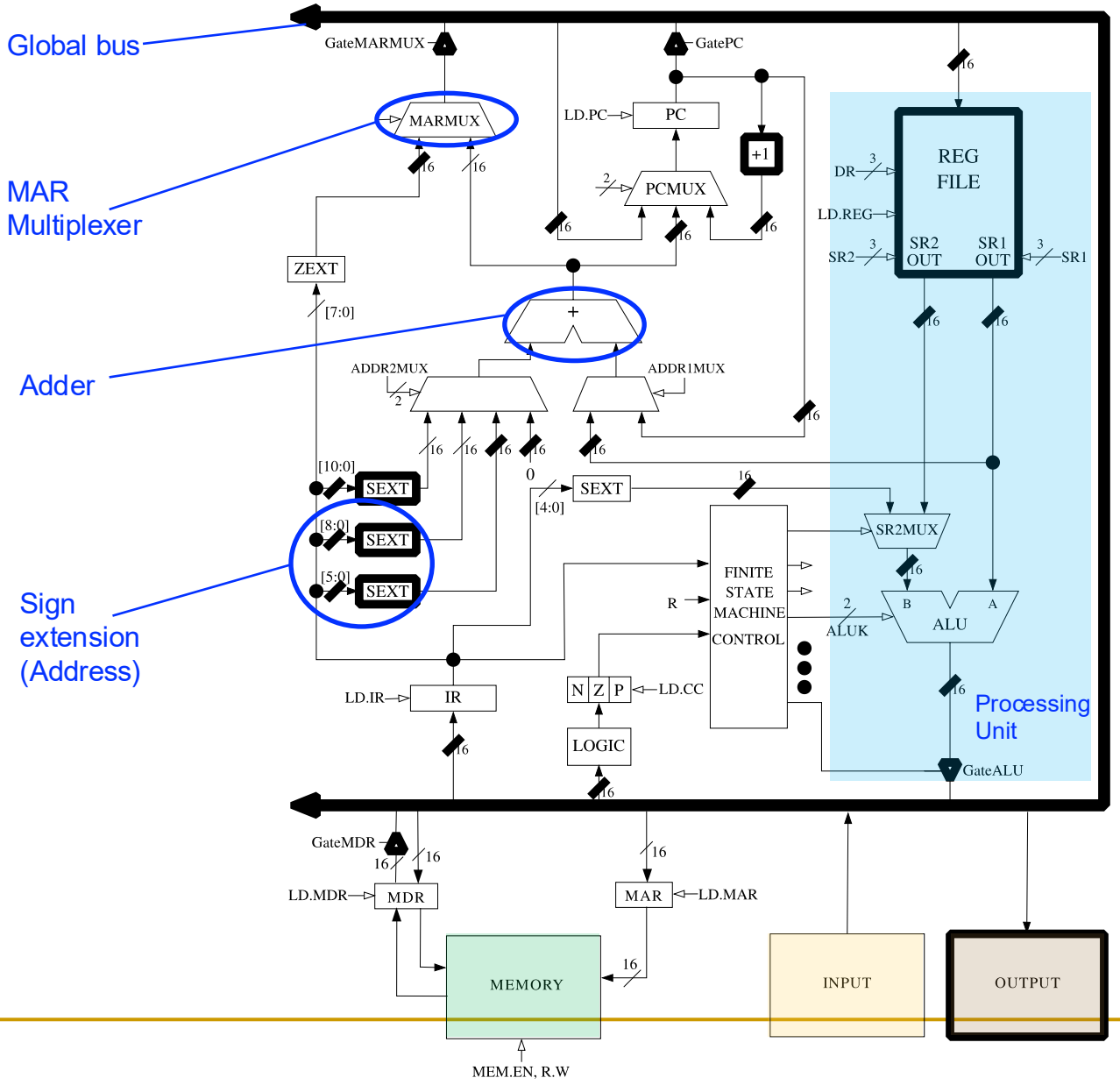
OP	DR	PCoffset9
E	5	0x1FD

Machine Code

OP	DR	PCoffset9
1 1 1 0	1 0 1	1 1 1 1 1 1 1 0 1
15 12	11 9	8 0



Address Calculation in LC-3 Data Path



Immediate Addressing Mode in MIPS

- In MIPS, lui (load upper immediate) loads a 16-bit immediate into the upper half of a register and sets the lower half to 0

- It is used to assign 32-bit constants to a register

High-level code

```
a = 0x6d5e4f3c;
```

MIPS assembly

```
# $s0 = a
lui  $s0, 0x6d5e
ori  $s0, 0x4f3c
```

Addressing Example in LC-3

- What is the final value of R3?

P&P, Chapter 5.3.5

Address	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
x30F6	1	1	1	0	0	0	1	1	1	1	1	1	1	1	0	1	R1<- PC-3
x30F7	0	0	0	1	0	1	0	0	0	1	1	0	1	1	1	0	R2<- R1+14
x30F8	0	0	1	1	0	1	0	1	1	1	1	1	1	0	1	1	M[x30F4]<- R2
x30F9	0	1	0	1	0	1	0	0	1	0	1	0	0	0	0	0	R2<- 0
x30FA	0	0	0	1	0	1	0	0	1	0	1	0	0	1	0	1	R2<- R2+5
x30FB	0	1	1	1	0	1	0	0	0	1	0	0	1	1	1	0	M[R1+14]<- R2
x30FC	1	0	1	0	0	1	1	1	1	1	1	1	0	1	1	1	R3<- M[M[x30FE]]

LC- ISA Instruction Encodings

Patt & Patel,
Back cover inside

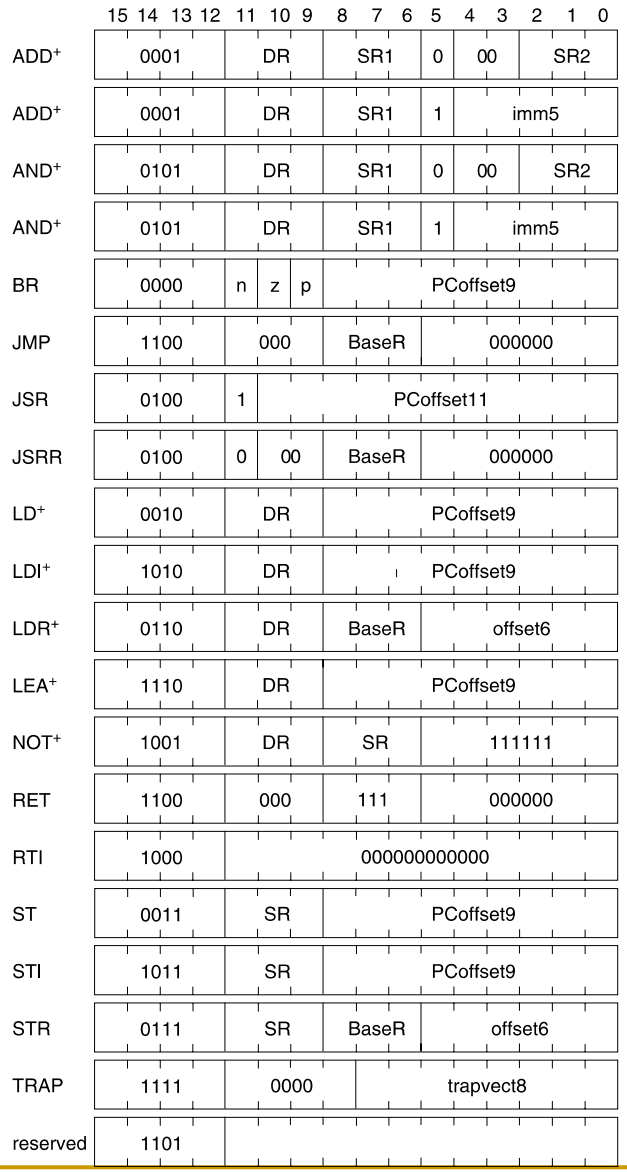


Figure 5.3 Formats of the entire LC-3 instruction set. NOTE: + indicates instructions that modify condition codes

Addressing Example in LC-3

- What is the final value of R3?

P&P, Chapter 5.3.5

Address	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
x30F6	LEA	1	1	1	0	0	0	1	-3								R1 = PC – 3 = 0x30F7 – 3 = 0x30F4
x30F7	ADD	0	0	1	0	1	0	0	0	1	1	14					R2 = R1 + 14 = 0x30F4 + 14 = 0x3102
x30F8	ST	0	1	1	0	1	0	-5									M[PC - 5] = M[0x030F4] = 0x3102
x30F9	AND	1	0	1	0	1	0	0	1	0	0	0	0	0			R2 = 0
x30FA	ADD	0	0	1	0	1	0	0	1	0	5						R2 = R2 + 5 = 5
x30FB	STR	1	1	1	0	1	0	0	0	1							M[R1 + 14] = M[0x30F4 + 14] = M[0x3102] = 5
x30FC	LDI	0	1	0	0	1	1	-9									R3 = M[M[PC – 9]] = M[M[0x30FD – 9]] = M[M[0x30F4]] = M[0x3102] = 5

- The final value of R3 is 5

Control Flow Instructions

Control Flow Instructions

- Allow a program to execute out of sequence
- Conditional branches and unconditional jumps
  - Conditional branches are used to make decisions
    - E.g., if-else statement
  - In LC-3, three condition codes are used
  - Jumps are used to implement
    - Loops
    - Function calls
  - JMP in LC-3 and j in MIPS
    - We have already seen these

Conditional Control Flow (Conditional Branching)

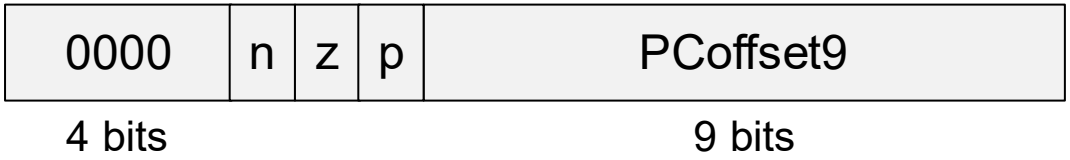
Condition Codes in LC-3

- Each time one GPR (R0-R7) is written, three single-bit registers are updated
- Each of these condition codes are either set (set to 1) or cleared (set to 0)
  - If the written value is negative
    - N is set, Z and P are cleared
  - If the written value is zero
    - Z is set, N and P are cleared
  - If the written value is positive
    - P is set, N and Z are cleared
- x86 and SPARC are examples of ISAs that use condition codes

Conditional Branches in LC-3

- BRz (Branch if Zero)

```
BRz   PCoffset9
```



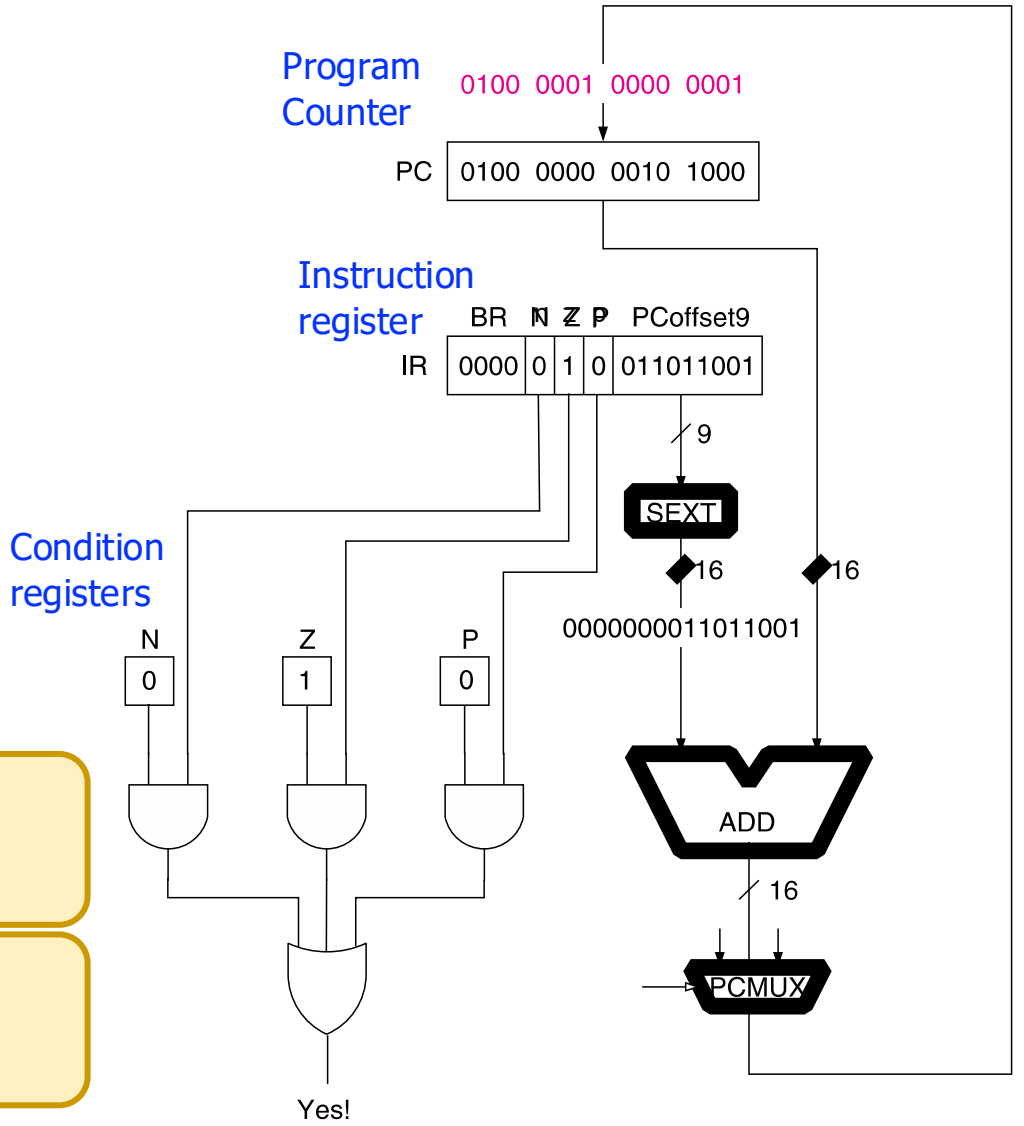
  - n, z, p = which condition code is tested (N, Z, and/or P)
    - n, z, p: instruction bits to identify the condition codes to be tested
    - N, Z, P: values of the corresponding condition codes
  - PCoffset9 = immediate or constant value
  - if ((n AND N) OR (p AND P) OR (z AND Z))
    - then PC ← PC† + sign-extend(PCoffset9)
  - Variations: BRn, BRz, BRp, BRzp, BRnp, BRnz, BRnzp

† This is the incremented PC

Conditional Branches in LC-3

- BRz

```
BRz   0x0D9
```



What if n = z = p = 1?*
(i.e., BRnzp)

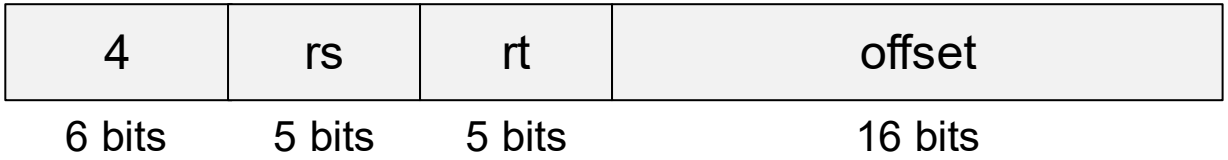
And what if n = z = p = 0?

*n, z, p are the instruction bits to identify the condition codes to be tested

Conditional Branches in MIPS

- beq (Branch if Equal)

```
beq  $s0, $s1, offset
```



  - 4 = opcode
  - rs, rt = source registers
  - offset = immediate or constant value
  - if rs == rt
    - then PC ← PC† + sign-extend(offset) * 4
  - Variations: beq, bne, blez, bgtz

† This is the incremented PC

Branch If Equal in MIPS and LC-3

MIPS assembly

```
beq  $s0, $s1, offset
```

LC-3 assembly

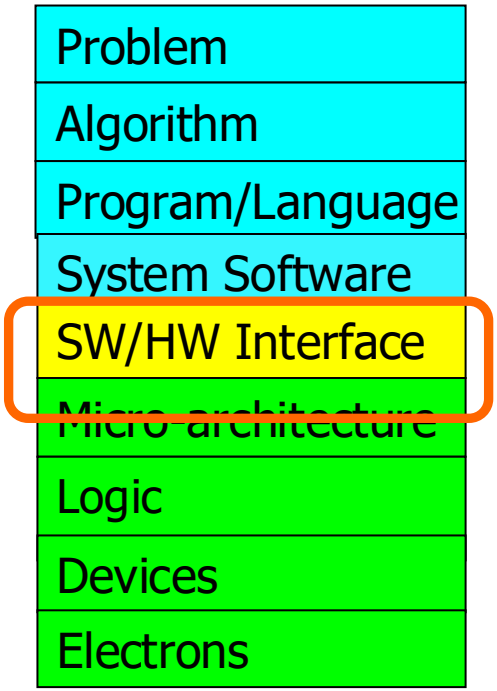
```
NOT  R2, R1
ADD  R3, R2, #1
ADD  R4, R3, R0
BRz  offset
```

Subtract (R0 - R1)

- This is an example of tradeoff in the instruction set
  - The same functionality requires more instructions in LC-3
  - But, the control logic requires more complexity in MIPS

What We Learned

- Basic elements of a computer & the von Neumann model
  - LC-3: An example von Neumann machine
- Instruction Set Architectures: LC-3 and MIPS
  - Operate instructions
  - Data movement instructions
  - Control instructions
- Instruction formats
- Addressing modes



There Is A Lot More to Cover on ISAs

https://www.youtube.com/onurmutlulectures

Many Different ISAs Over Decades

- x86
- PDP-x: Programmed Data Processor (PDP-11)
- VAX
- IBM 360
- CDC 6600
- SIMD ISAs: CRAY-1, Connection Machine
- VLIW ISAs: Multiflow, Cydrome, IA-64 (EPIC)
- PowerPC, POWER
- RISC ISAs: Alpha, MIPS, SPARC, ARM, RISC-V, …

- What are the fundamental differences?
  - E.g., how instructions are specified and what they do
  - E.g., how complex are instructions, data types, addr. modes

Complex vs. Simple Instructions+Data Types

- Complex instruction: An instruction does a lot of work, e.g. many operations
  - Insert in a doubly linked list
  - Compute FFT
  - String copy
  - Matrix multiply
  - …
- Simple instruction: An instruction does little work -- it is a primitive using which complex operations can be built
  - Add
  - XOR
  - Multiply
  - …

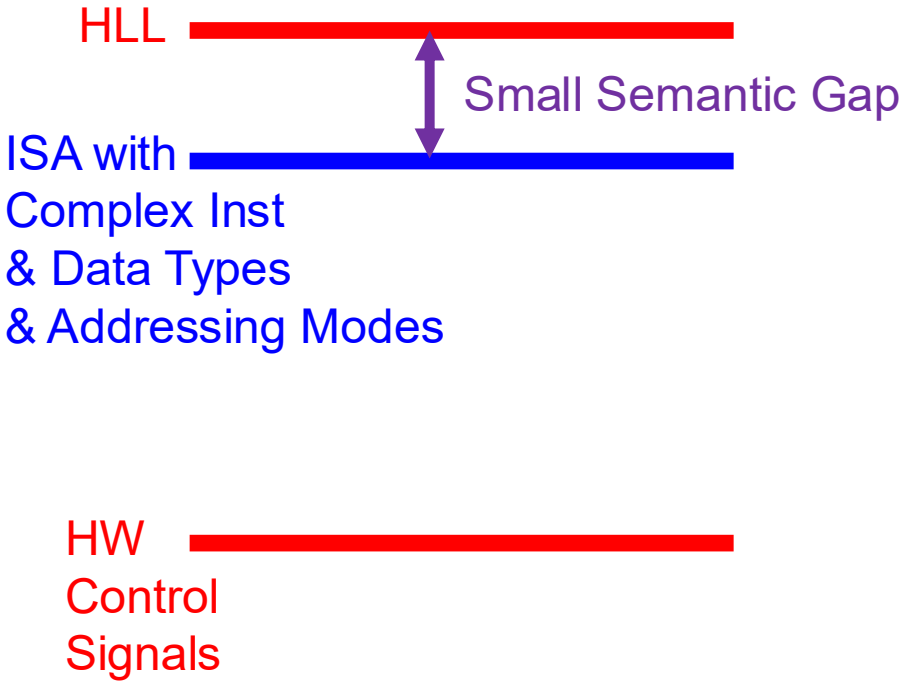
Complex vs. Simple Instructions+Data Types

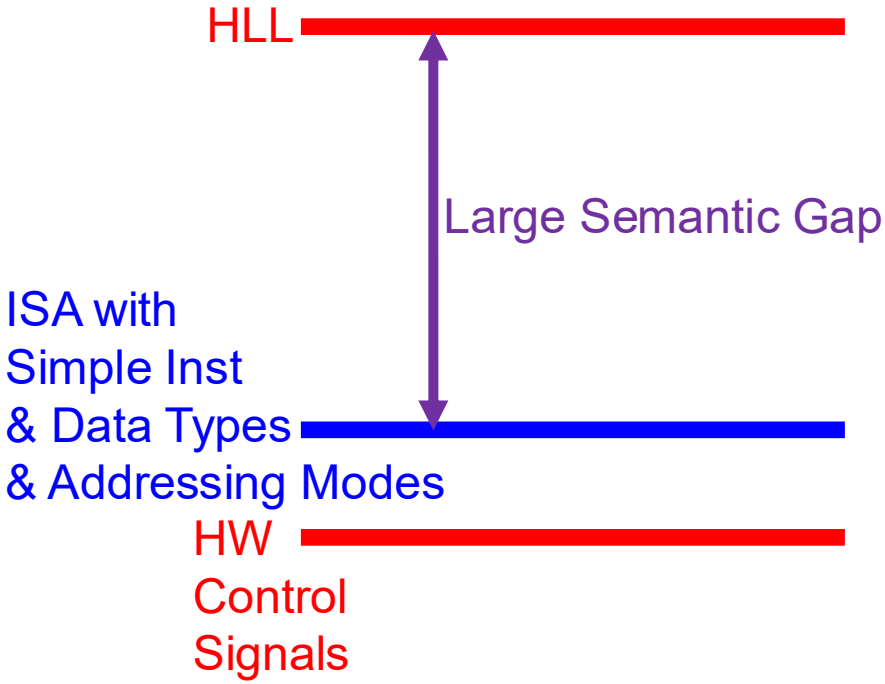
- Advantages of Complex Instructions + Data Types
  - + Denser encoding → smaller code size → better memory utilization, saves off-chip bandwidth, better cache hit rate (better packing of instructions)
  - + Simpler compiler: no need to optimize small instructions as much

- Disadvantages of Complex Instructions + Data Types
  - - Larger chunks of work → compiler has less opportunity to optimize (limited in fine-grained optimizations it can do)
  - - More complex hardware → translation from a high level to control signals and optimization needs to be done by hardware

Semantic Gap

- How close instructions & data types & addressing modes are to high-level language (HLL)





Easier mapping of HLL to ISA
Less work for software designer
More work for hardware designer
Optimization burden on HW

Harder mapping of HLL to ISA
More work for software designer
Less work for hardware designer
Optimization burden on SW

We Covered Until Here in Lecture

Digital Design & Computer Arch.

Lecture 8: Instruction Set Architectures II

Prof. Onur Mutlu

ETH Zürich

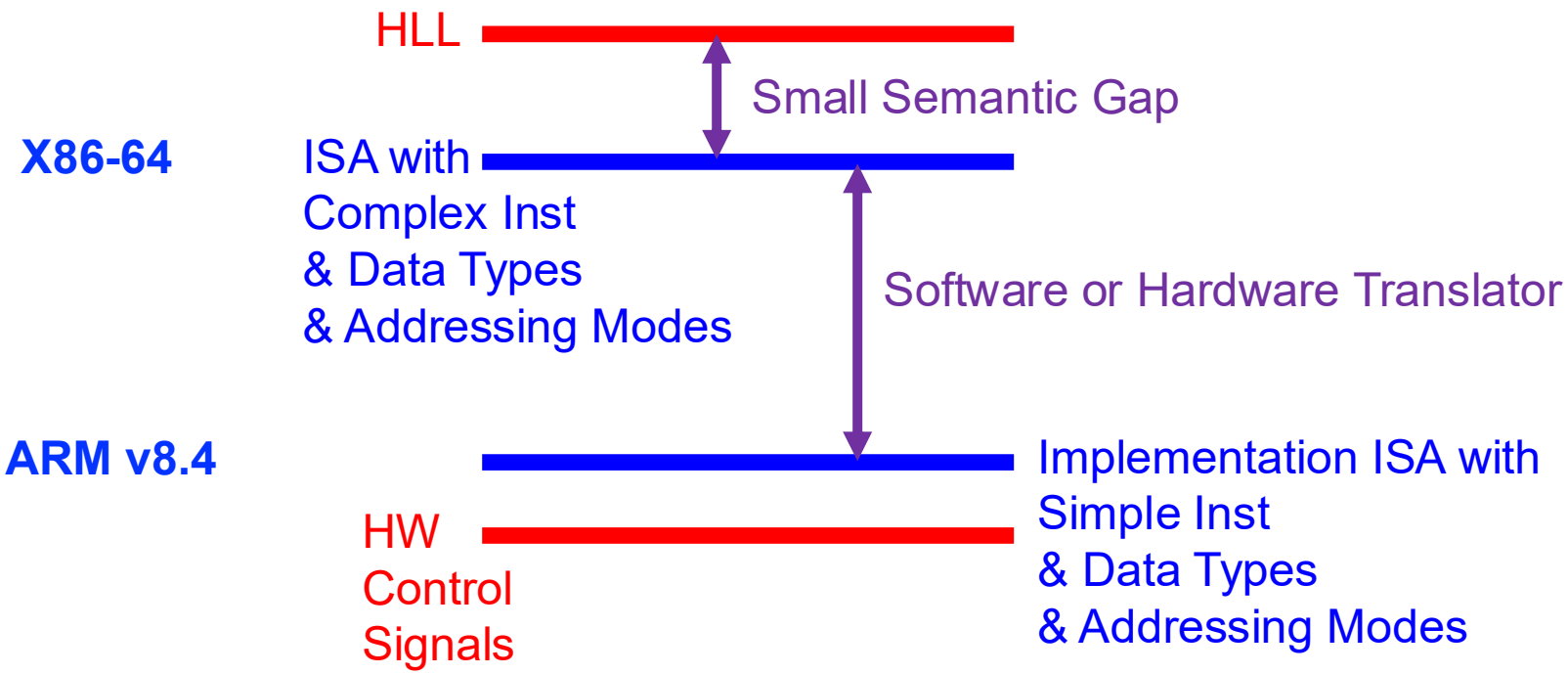
Spring 2026

13 March 2026

Further Slides for Your Own Study (May Be Covered in Future Lectures)

How to Change the Semantic Gap Tradeoffs

- Translate from one ISA into a different "implementation" ISA



An Example: Rosetta 2 Binary Translator

Rosetta 2 [edit]

In 2020, Apple announced Rosetta 2 would be bundled with macOS Big Sur, to aid in the Mac transition to Apple silicon. The software permits many applications compiled exclusively for execution on x86-64-based processors to be translated for execution on Apple silicon.[2][8]

In addition to the just-in-time (JIT) translation support, Rosetta 2 offers ahead-of-time compilation (AOT), with the x86-64 code fully translated, just once, when an application without a universal binary is installed on an Apple silicon Mac.[9]

Rosetta 2's performance has been praised greatly.[10][11] In some benchmarks, x86-64-only programs performed better under Rosetta 2 on a Mac with an Apple M1 SOC than natively on a Mac with an Intel x86-64 processor. One of the key reasons why Rosetta 2 provides such high level of translation efficiency is the support of x86-64 memory ordering in Apple M1 SOC.[12]

Mac transition to Apple silicon

Apple silicon · ARM architecture · Universal 2 binary · Rosetta 2 · Developer Transition Kit

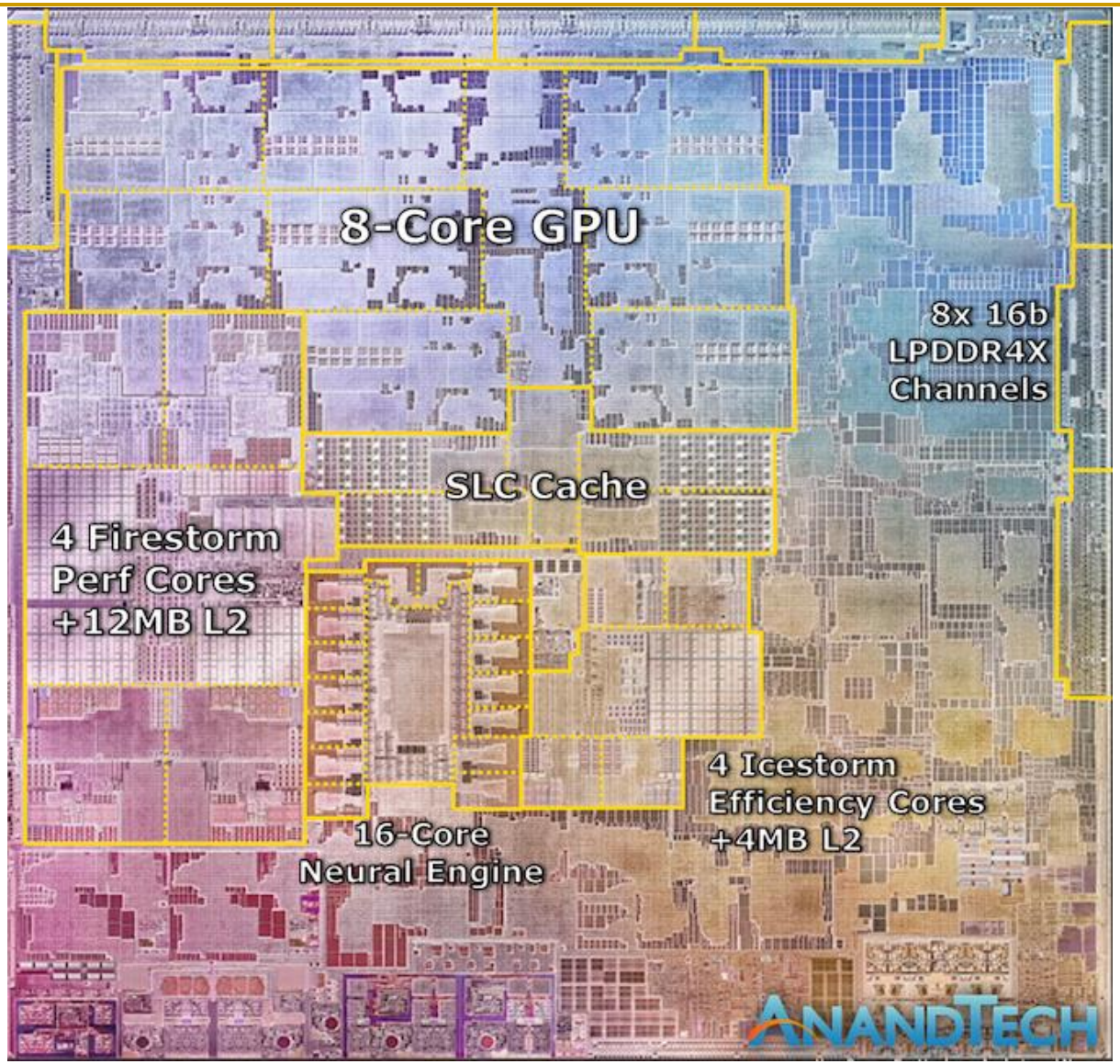
V · T · E

Although Rosetta 2 works for most software, some software doesn't work at all[13] or is reported to be "sluggish".[14] A lot of software can be made compatible with the new Macs by the vendor recompiling the software, often a simple task; while for some software (such as software that includes assembly language code, or that generates machine code), the changes to make them work aren't simple and cannot be automated.

Similar to the first version, Rosetta 2 does not normally require user intervention. When a user attempts to launch an x86-64-only application for the first time, macOS prompts them to install Rosetta 2 if it is not already available. Subsequent launches of x86-64 programs will execute via translation automatically. An option also exists to force a universal binary to run as x86-64 code through Rosetta 2, even on an ARM-based machine.[15]

https://en.wikipedia.org/wiki/Rosetta_(software)#Rosetta_2

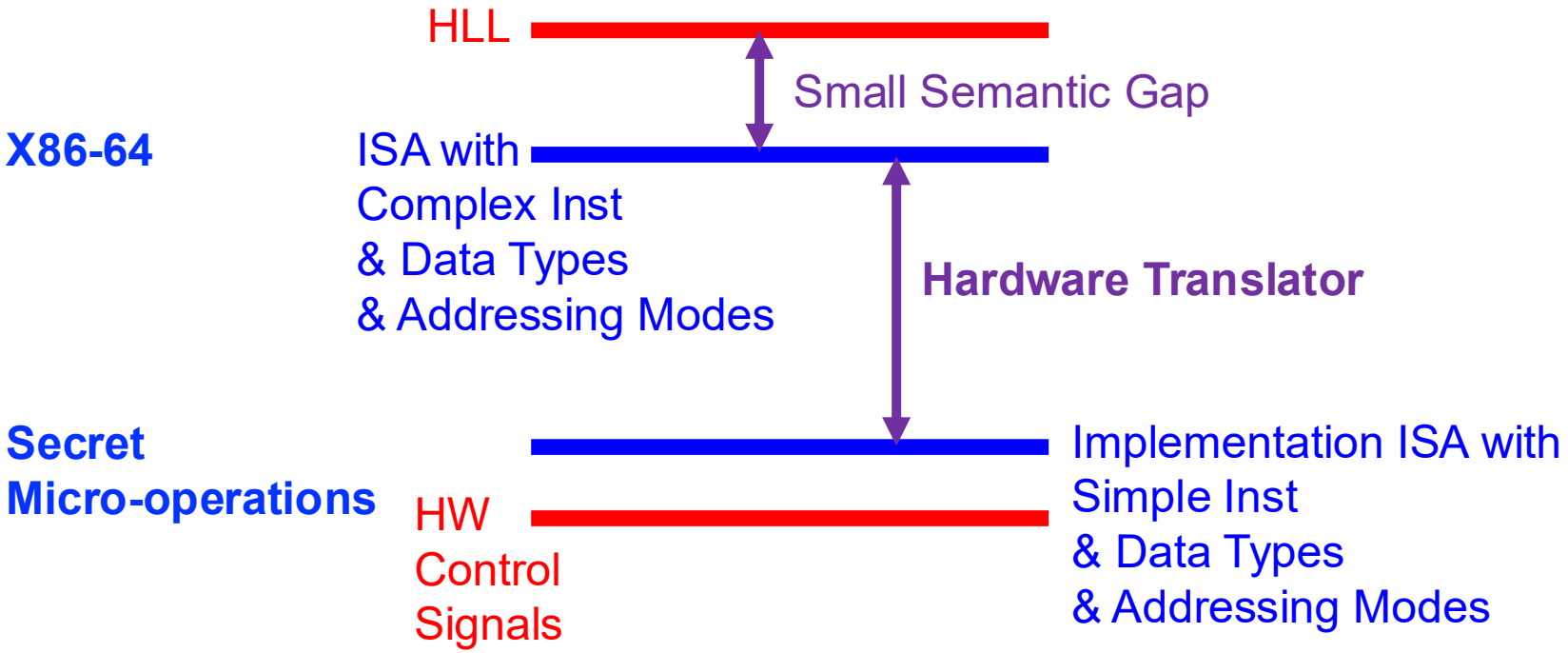
An Example: Rosetta 2 Binary Translator



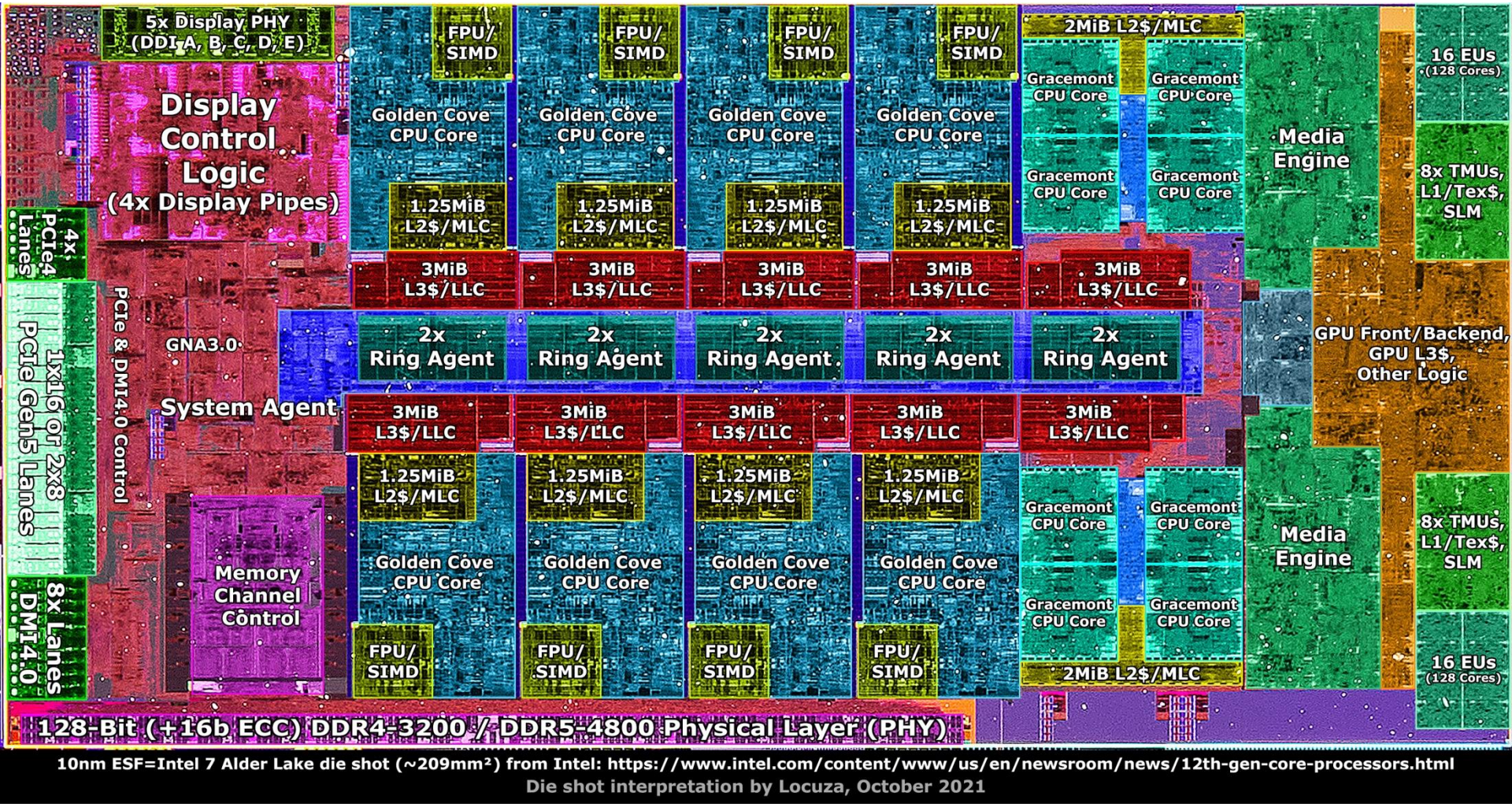
Apple M1, 2021

Source: https://www.anandtech.com/show/16252/mac-mini-apple-m1-tested

Another Example: Intel and AMD Processors



Another Example: Intel and AMD Processors

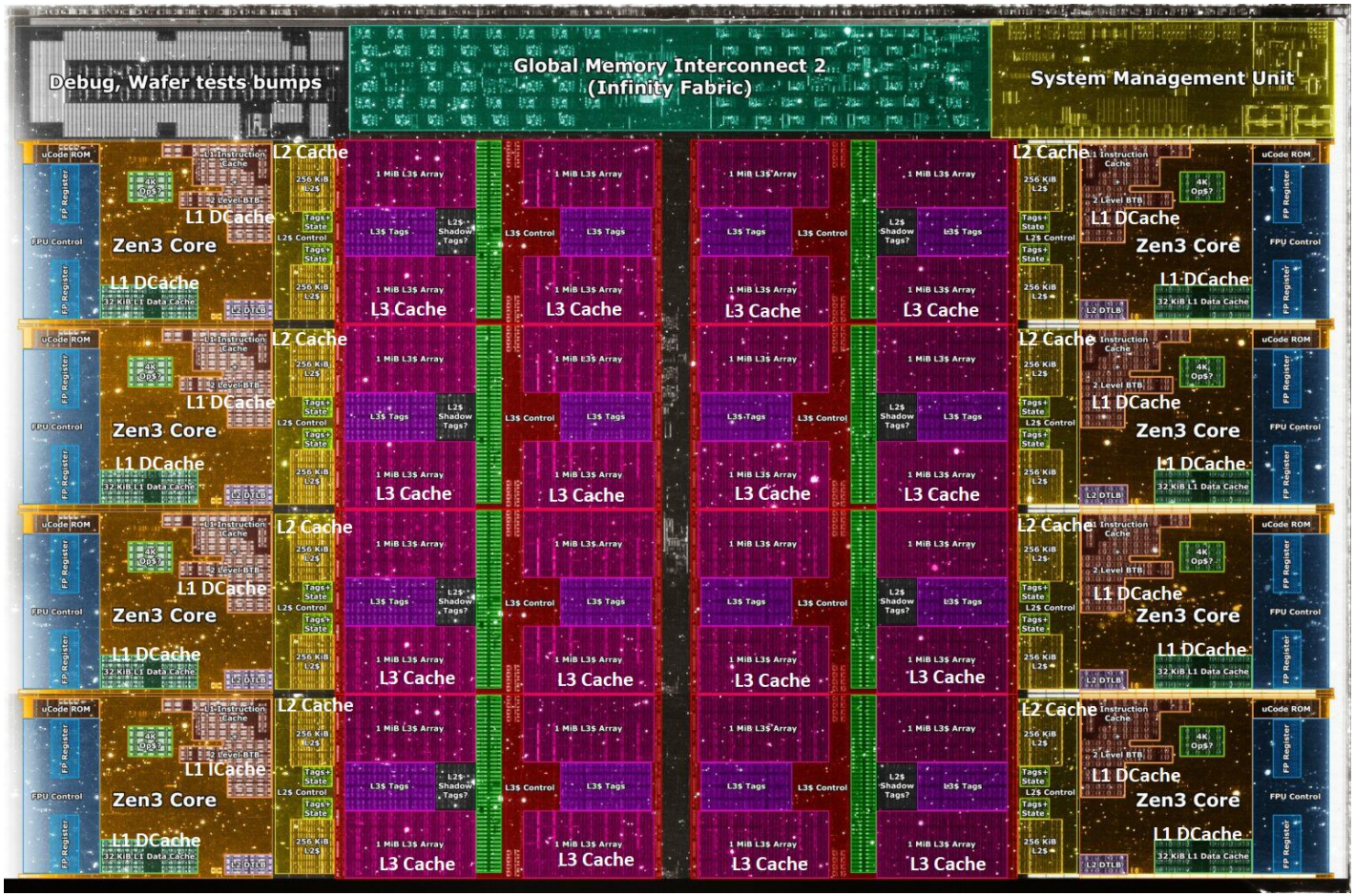


10nm ESF=Intel 7 Alder Lake die shot (~209mm²) from Intel: https://www.intel.com/content/www/us/en/newsroom/news/12th-gen-core-processors.html
Die shot interpretation by Locuza, October 2021

Intel Alder Lake, 2021

Source: https://twitter.com/Locuza_/status/1454152714930331652

Another Example: Intel and AMD Processors



AMD Ryzen 5000, 2020

Core Count:
8 cores/16 threads

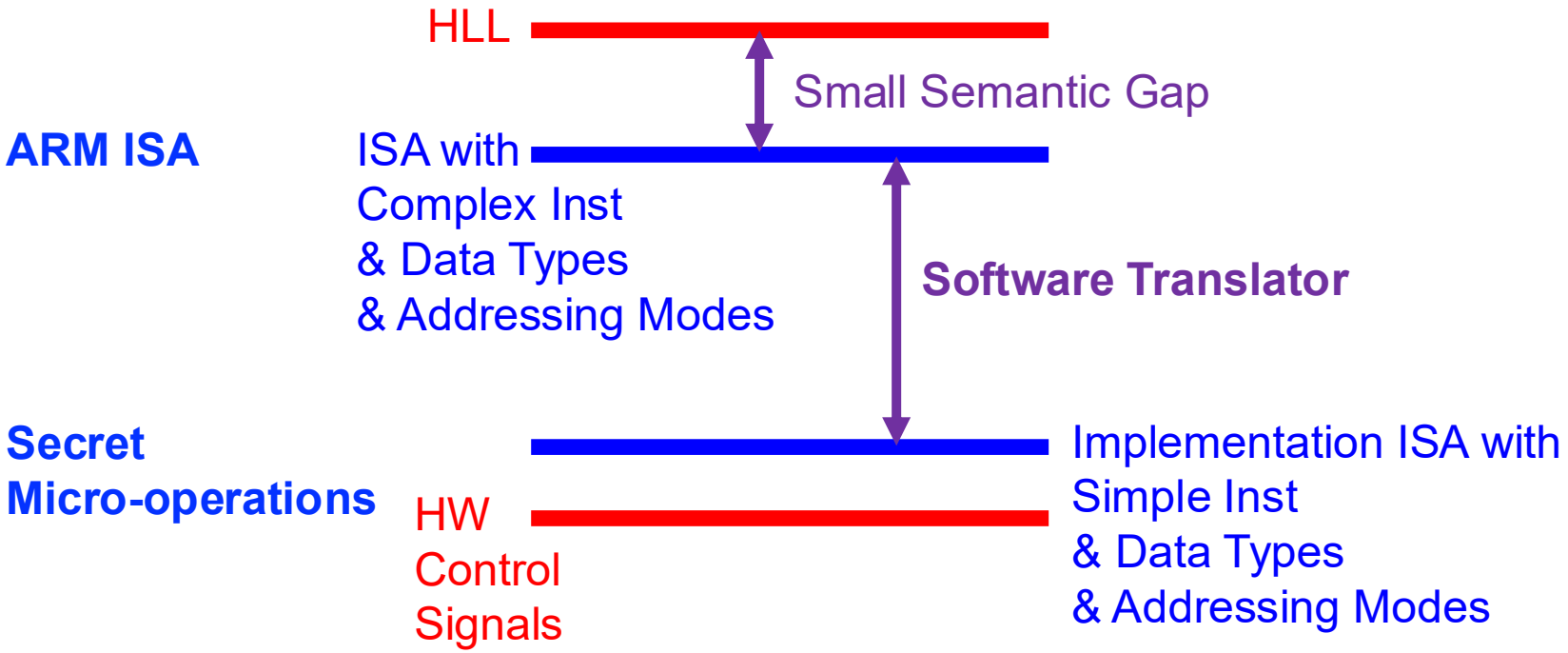
L1 Caches:
32 KB per core

L2 Caches:
512 KB per core

L3 Cache:
32 MB shared

https://wccftech.com/amd-ryzen-5000-zen-3-vermeer-undressed-high-res-die-shots-close-ups-pictured-detailed/

Another Example: NVIDIA Denver

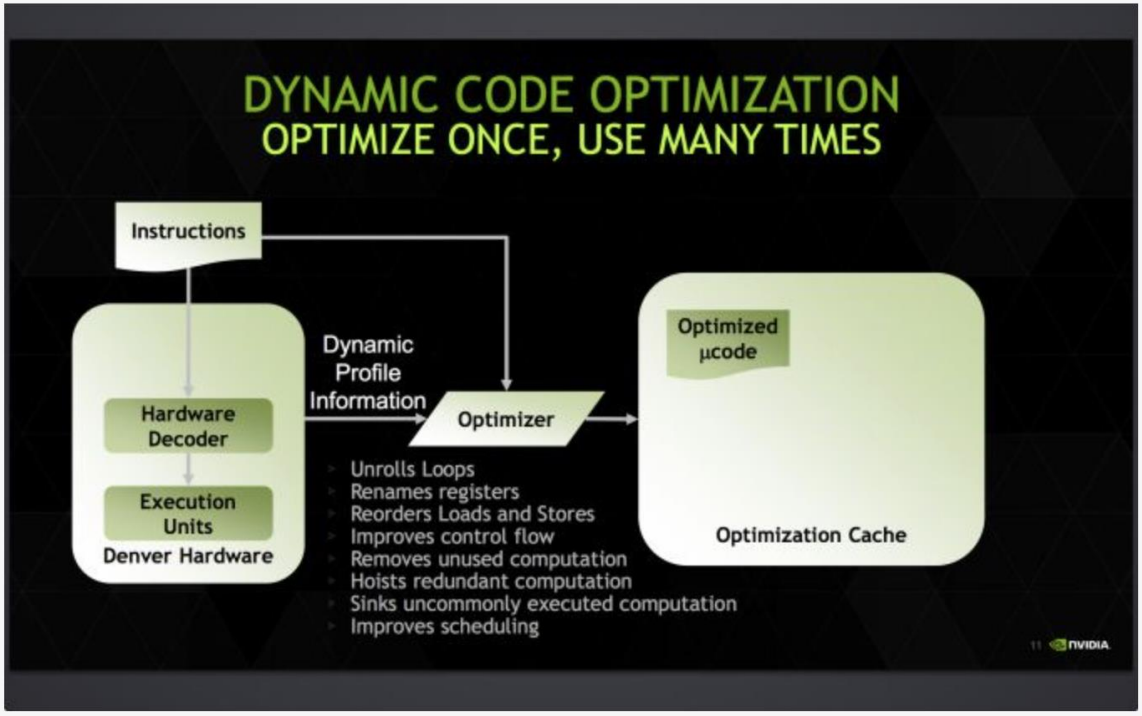


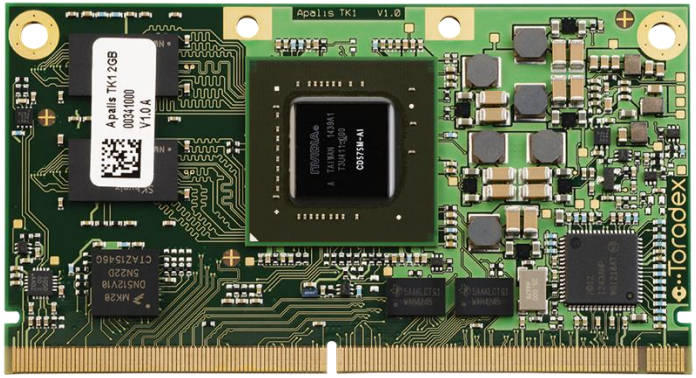
Another Example: NVIDIA Denver

The Secret of Denver: Binary Translation & Code Optimization

As we alluded to earlier, NVIDIA's decision to forgo a traditional out-of-order design for Denver means that much of Denver's potential is contained in its software rather than its hardware. The underlying chip itself, though by no means simple, is at its core a very large in-order processor. So it falls to the software stack to make Denver sing.

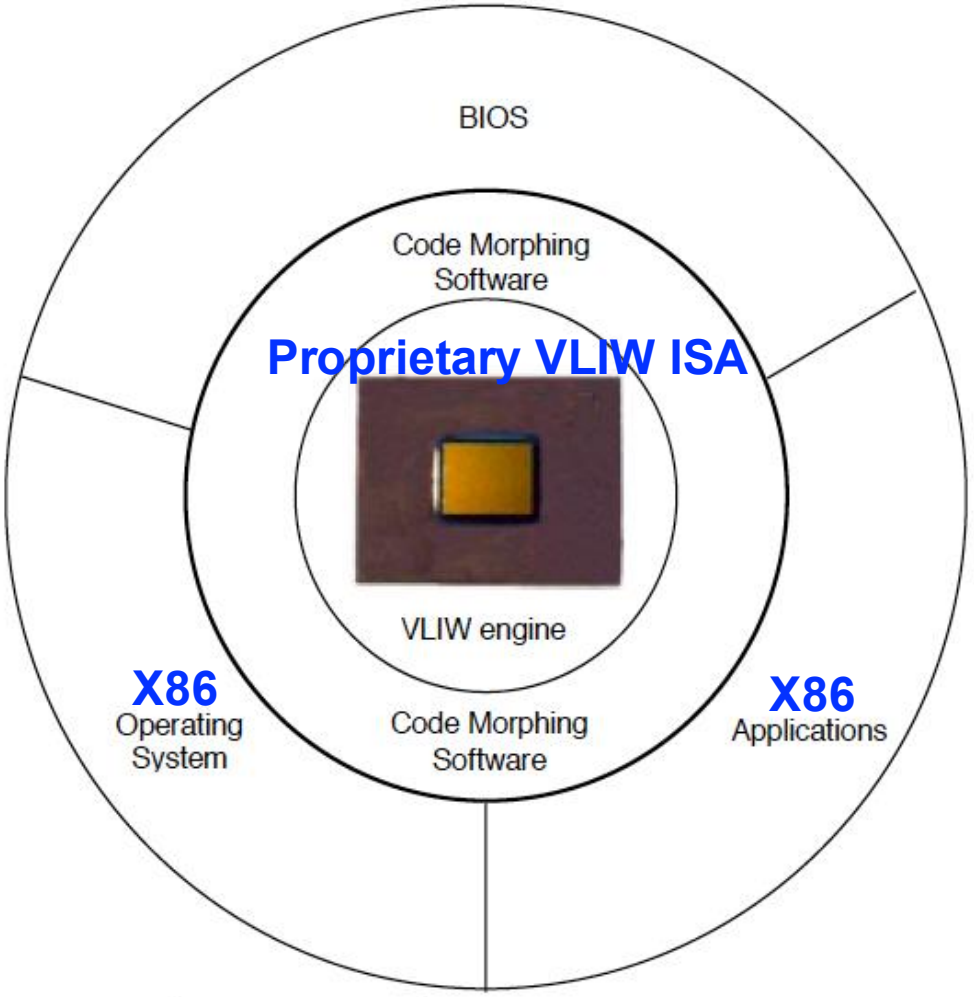
Accomplishing this task is NVIDIA's dynamic code optimizer (DCO). The purpose of the DCO is to accomplish two tasks: to translate ARM code to Denver's native format, and to optimize this code to make it run better on Denver. With no out-of-order hardware on Denver, it is the DCO's task to find instruction level parallelism within a thread to fill Denver's many execution units, and to reorder instructions around potential stalls, something that is no simple task.





https://www.anandtech.com/show/8701/the-google-nexus-9-review/4
https://www.toradex.com/computer-on-modules/apalis-arm-family/nvidia-tegra-k1

Transmeta: x86 to VLIW Translation



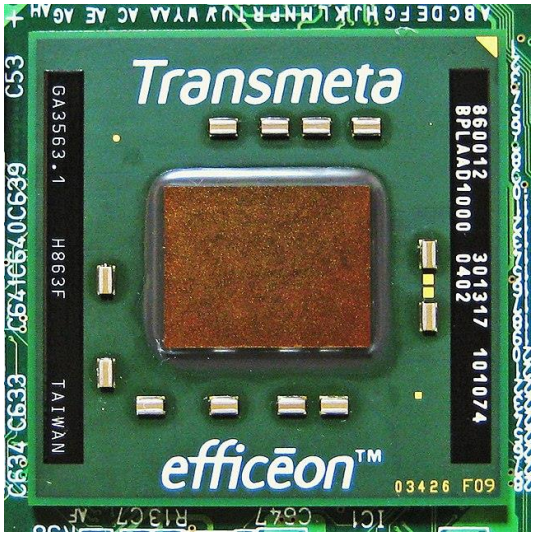


Figure 5. The Code Morphing software mediates between x86 software and the Crusoe processor.

Klaiber, "The Technology Behind Crusoe Processors," Transmeta White Paper 2000.

https://www.wikiwand.com/en/Transmeta_Efficeon

ISA-level Tradeoffs: Number of Registers

- Affects:
  - Number of bits used for encoding register address
  - Number of values kept in fast storage (register file)
  - (uarch) Size, access time, power consumption of register file

- Large number of registers:

  + Enables better register allocation (and optimizations) by compiler → fewer saves/restores

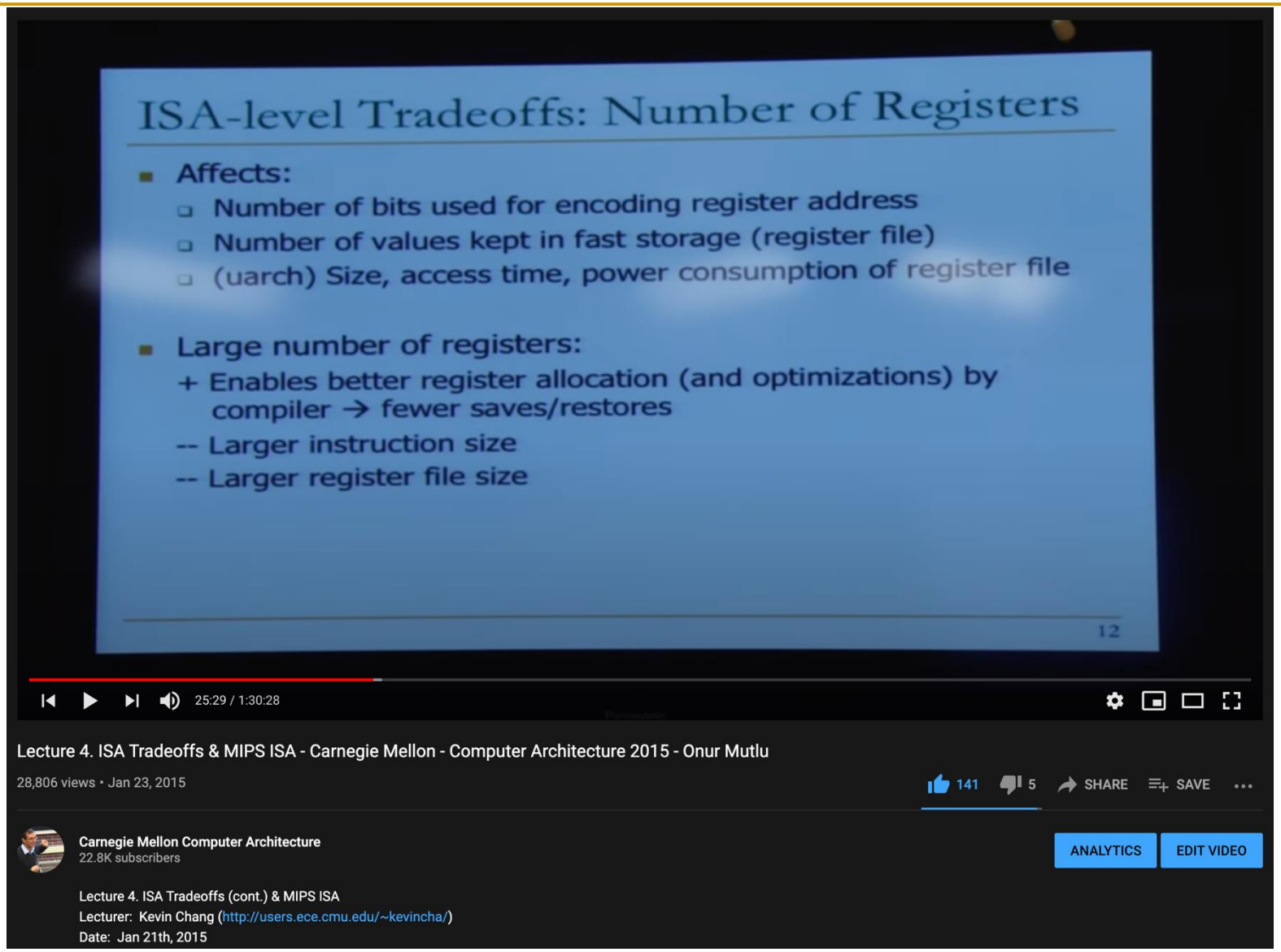
  -- Larger instruction size

  -- Larger register file size

There Is A Lot More to Cover on ISAs

https://www.youtube.com/onurmutlulectures

There Is A Lot More to Cover on ISAs



https://www.youtube.com/onurmutlulectures

Detailed Lectures on ISAs & ISA Tradeoffs

- Computer Architecture, Spring 2015, Lecture 3
  - ISA Tradeoffs (CMU, Spring 2015)
  - https://www.youtube.com/watch?v=QKdiZSfwg-g&list=PL5PHm2jkkXmi5CxxI7b3JCL1TWybTDtKq&index=3
- Computer Architecture, Spring 2015, Lecture 4
  - ISA Tradeoffs & MIPS ISA (CMU, Spring 2015)
  - https://www.youtube.com/watch?v=RBgeCCW5Hjs&list=PL5PHm2jkkXmi5CxxI7b3JCL1TWybTDtKq&index=4
- Computer Architecture, Spring 2015, Lecture 2
  - Fundamental Concepts and ISA (CMU, Spring 2015)
  - https://www.youtube.com/watch?v=NpC39uS4K4o&list=PL5PHm2jkkXmi5CxxI7b3JCL1TWybTDtKq&index=2

ISA Design and Tradeoffs: More Critical Thinking

The Von Neumann Model/Architecture

Stored program

Sequential instruction processing

The von Neumann Model/Architecture

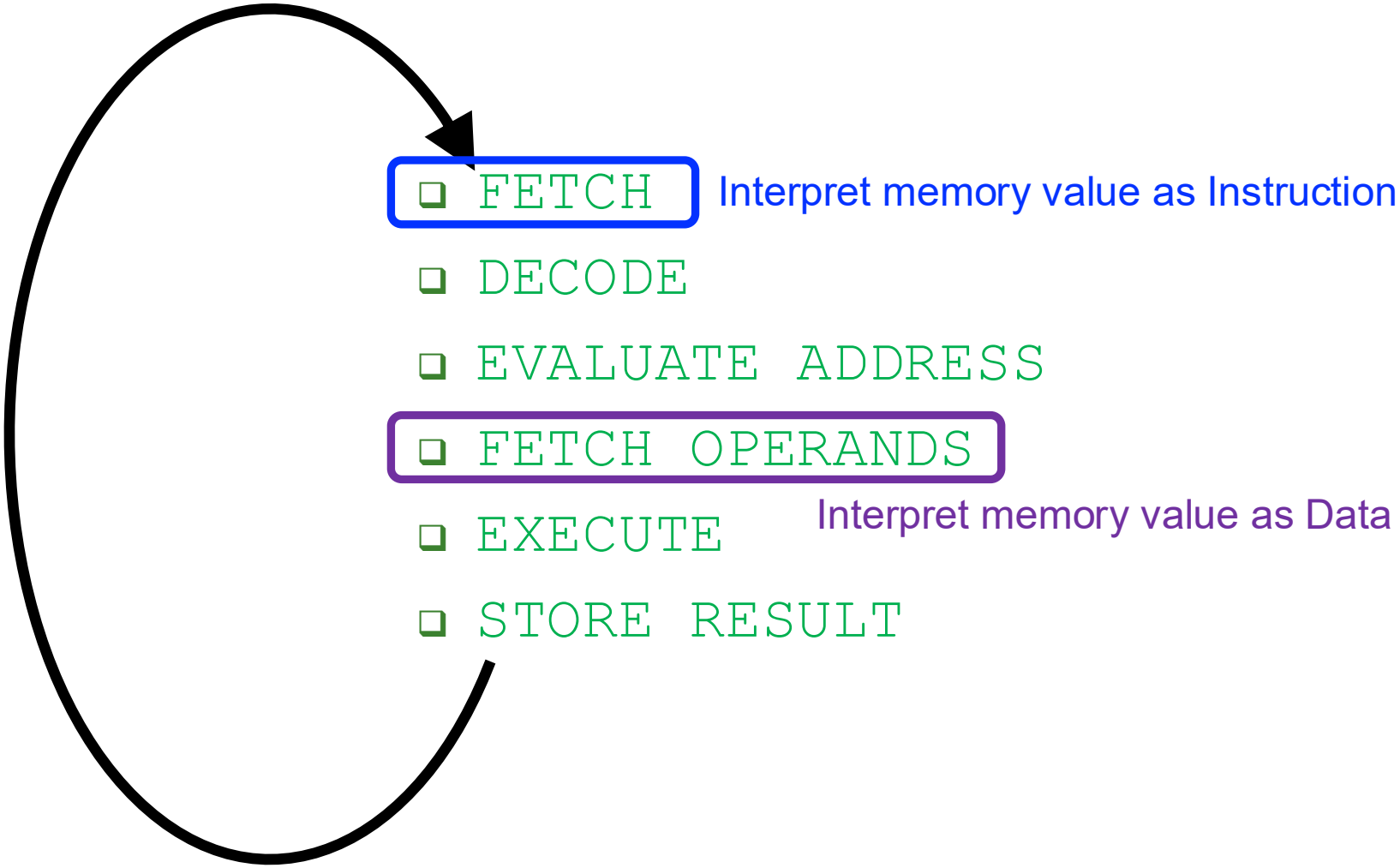
- Von Neumann model is also called *stored program computer* (instructions in memory). It has two key properties:

- Stored program
  - Instructions stored in a linear memory array
  - Memory is unified between instructions and data
    - The interpretation of a stored value depends on the control signals

      When is a value interpreted as an instruction?

- Sequential instruction processing

Recall: The Instruction Cycle



Whether a value fetched from memory is interpreted as an instruction depends on **when** that value is **fetched** in the instruction processing cycle.

The von Neumann Model/Architecture

- Von Neumann model is also called *stored program computer* (instructions in memory). It has two key properties:

- Stored program
  - Instructions stored in a linear memory array
  - Memory is unified between instructions and data
    - The interpretation of a stored value depends on the control signals

When is a value interpreted as an instruction?

- Sequential instruction processing
  - One instruction processed (fetched, executed, completed) at a time
  - Program counter (instruction pointer) identifies the current instruction
  - Program counter is advanced sequentially except for control transfer instructions

The von Neumann Model/Architecture

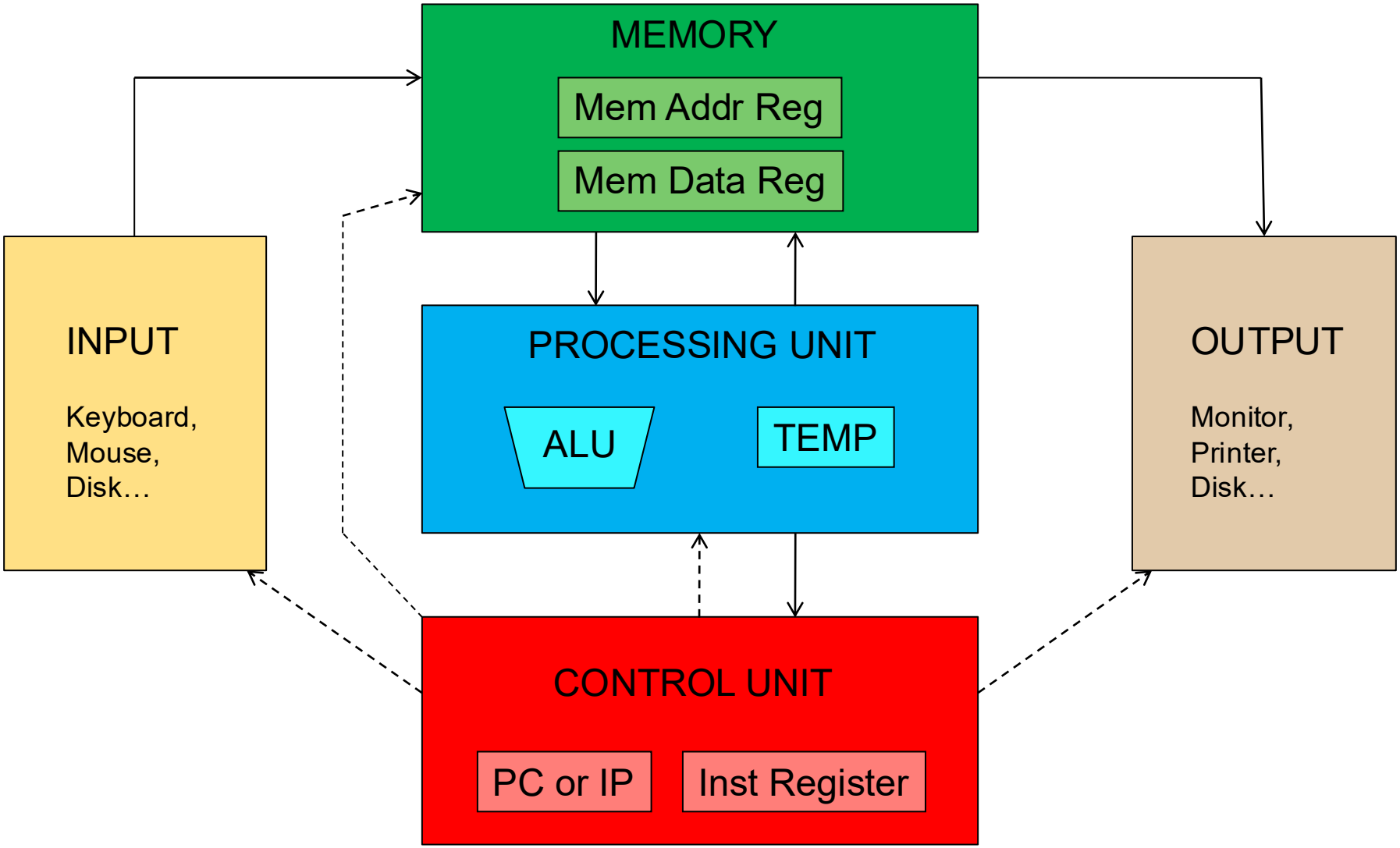
- Recommended reading
  - Burks, Goldstein, von Neumann, "Preliminary discussion of the logical design of an electronic computing instrument," 1946.

- Important reading
  - Patt and Patel book, Chapter 4, "The von Neumann Model"

- **Stored program**

- **Sequential instruction processing**

The Von Neumann Model (of a Computer)



The Von Neumann Model (of a Computer)

- Q: Is this the only way that a computer can process computer programs?

The von Neumann Model

- In order to build a computer, we need an execution model for processing computer programs
- John von Neumann proposed a fundamental model in 1946

- The von Neumann Model consists of 5 components
  - Memory (stores the program and data)
  - Processing unit
  - Input
  - Output
  - Control unit (controls the order in which instructions are carried out)
- Throughout this lecture, we will examine two examples of the von Neumann model
  - LC-3
  - MIPS

Burks, Goldstein, von Neumann, "Preliminary discussion of the logical design of an electronic computing instrument," 1946.

All general-purpose computers today use the von Neumann model

14

- A: No.
- Qualified Answer: No. But, it has been the dominant way
  - i.e., the dominant paradigm for computing
  - for N decades

Let's examine a **completely different model** for processing computer programs

The Dataflow Execution Model of a Computer

The Dataflow Model (of a Computer)

- Von Neumann model: An instruction is fetched and executed in control flow order
  - As specified by the program counter (instruction pointer)
  - Sequential unless explicit control flow instruction

- Dataflow model: An instruction is fetched and executed in data flow order
  - i.e., when its operands are ready
  - i.e., there is no program counter (instruction pointer)
  - Instruction ordering specified by data flow dependence
    - Each instruction specifies "who" should receive the result
    - An instruction can "fire" whenever all operands are received
  - Potentially many instructions can execute at the same time
    - Inherently more parallel

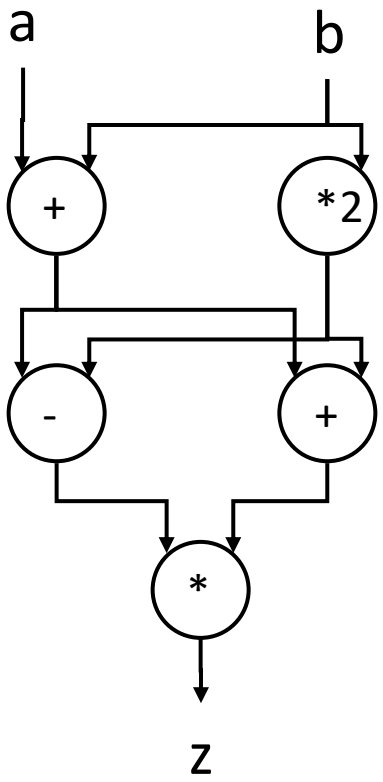
Von Neumann vs. Dataflow

- Consider a Von Neumann program
  - What is the significance of the program order?
  - What is the significance of the storage locations?

```
v = a + b;
w = b * 2;
x = v - w
y = v + w
z = x * y
```

Sequential

a, b are the only inputs
z is the only output



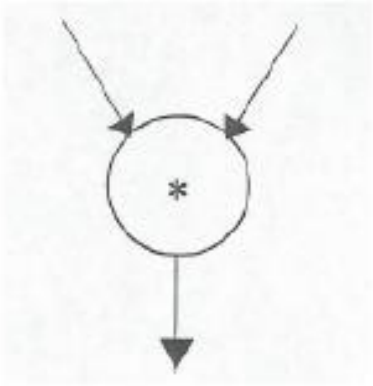
Dataflow

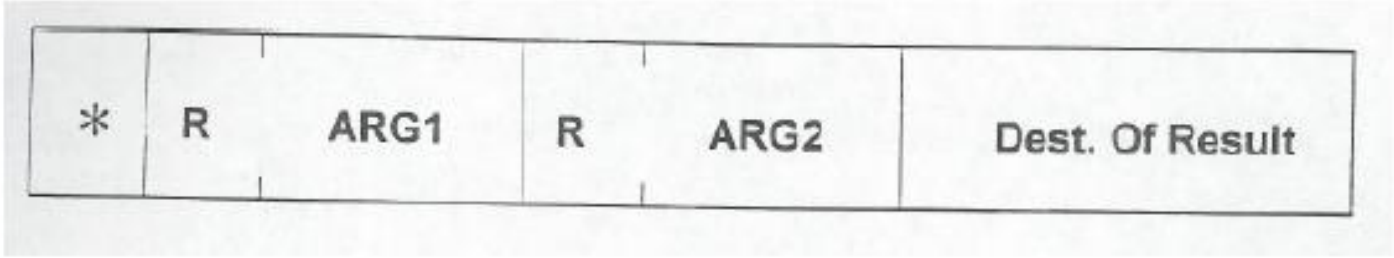
Which model is more natural to you as a programmer?

More on Dataflow

- In a dataflow machine, a program consists of dataflow nodes
  - A dataflow node fires (fetched and executed) when all it inputs are ready
    - i.e. when all inputs have tokens

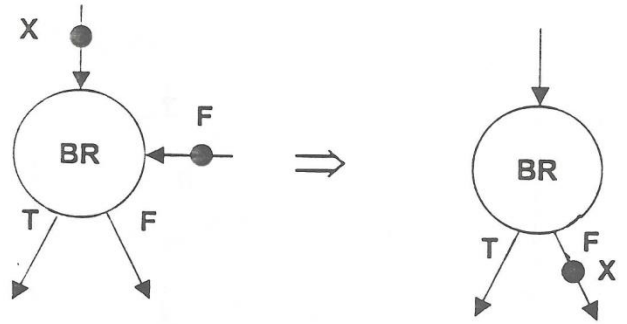
- Dataflow node and its ISA representation



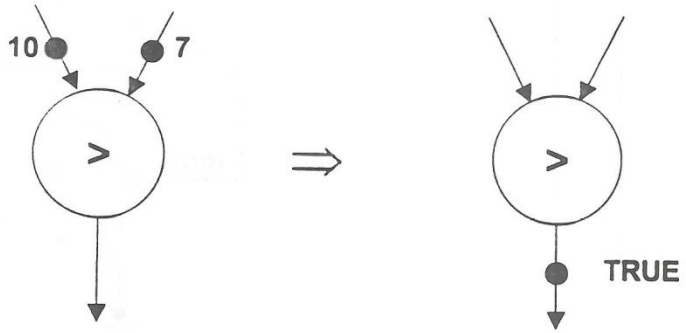


Example Dataflow Nodes

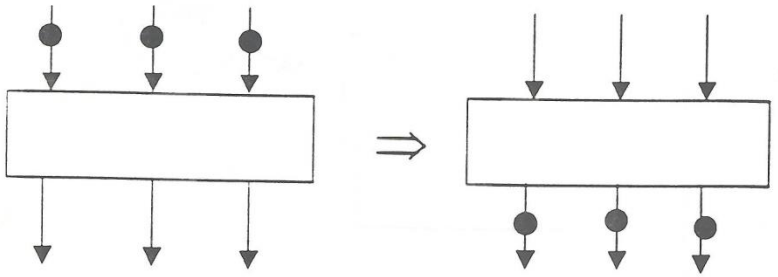
*Conditional



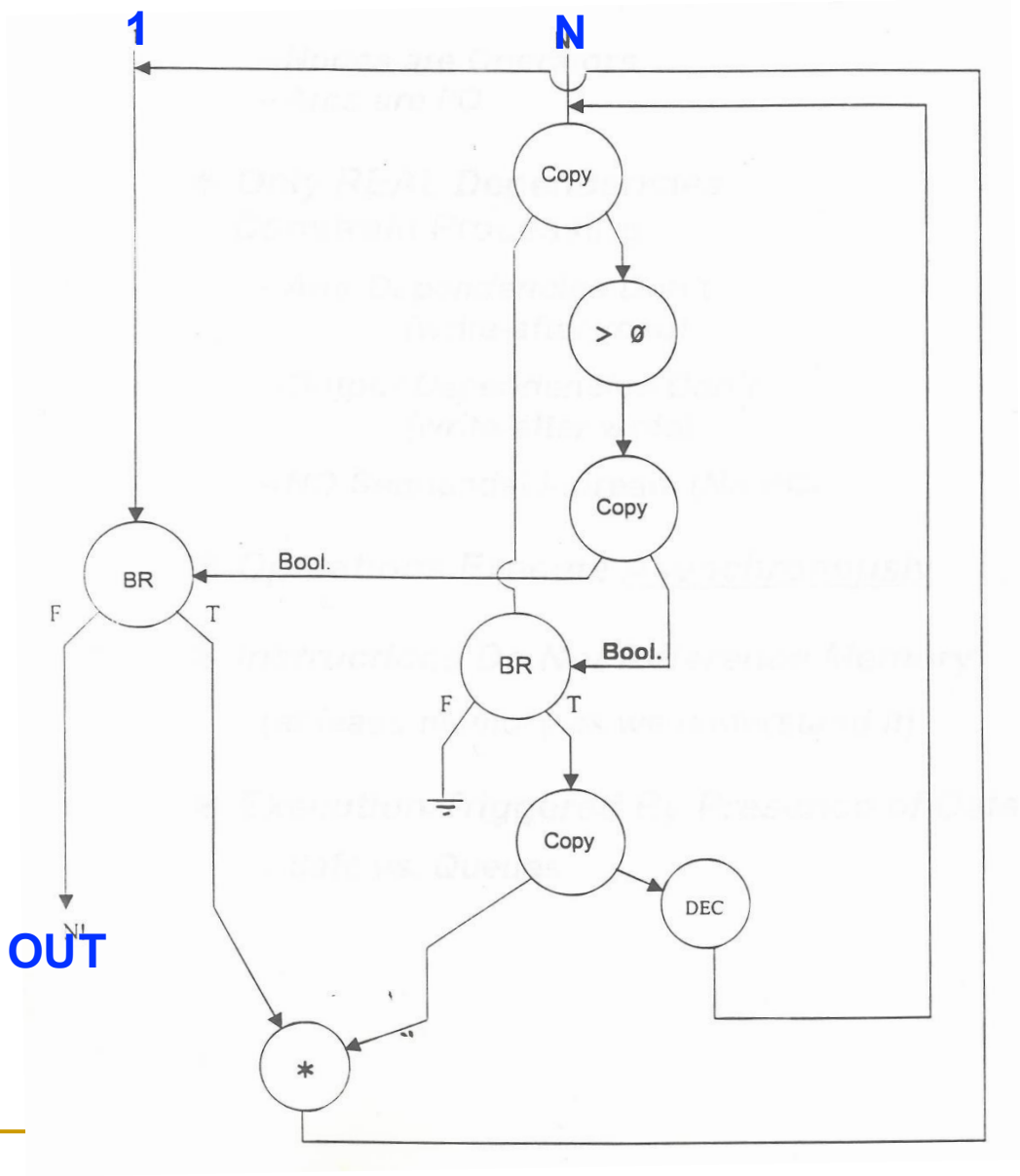
*Relational



*Barrier Synch



A Simple Example Dataflow Program



N is a non-negative integer

What is the value of OUT?

ISA-level Tradeoff: Program Counter

- Do we want a Program Counter (PC or IP) in the ISA?
  - Yes: Control-driven, sequential execution
    - An instruction is executed when the PC points to it
    - PC automatically changes sequentially (except for control flow instructions) → sequential
  - No: Data-driven, parallel execution
    - An instruction is executed when all its operand values are available → dataflow

- Tradeoffs: MANY high-level ones
  - Ease of programming (for average programmers)?
  - Ease of compilation?
  - Performance: Extraction of parallelism?
  - Hardware complexity?

ISA vs. Microarchitecture Level Tradeoff

- A similar tradeoff (control vs. data-driven execution) can be made at the microarchitecture level

- ISA: Specifies how the **programmer sees** the instructions to be executed
  - Programmer sees a sequential, control-flow execution order vs.
  - Programmer sees a dataflow execution order

- Microarchitecture: How the **underlying implementation actually executes** instructions
  - Microarchitecture can execute instructions in any order as long as it obeys the semantics specified by the ISA when making the instruction results visible to software
    - Programmer should see the order specified by the ISA

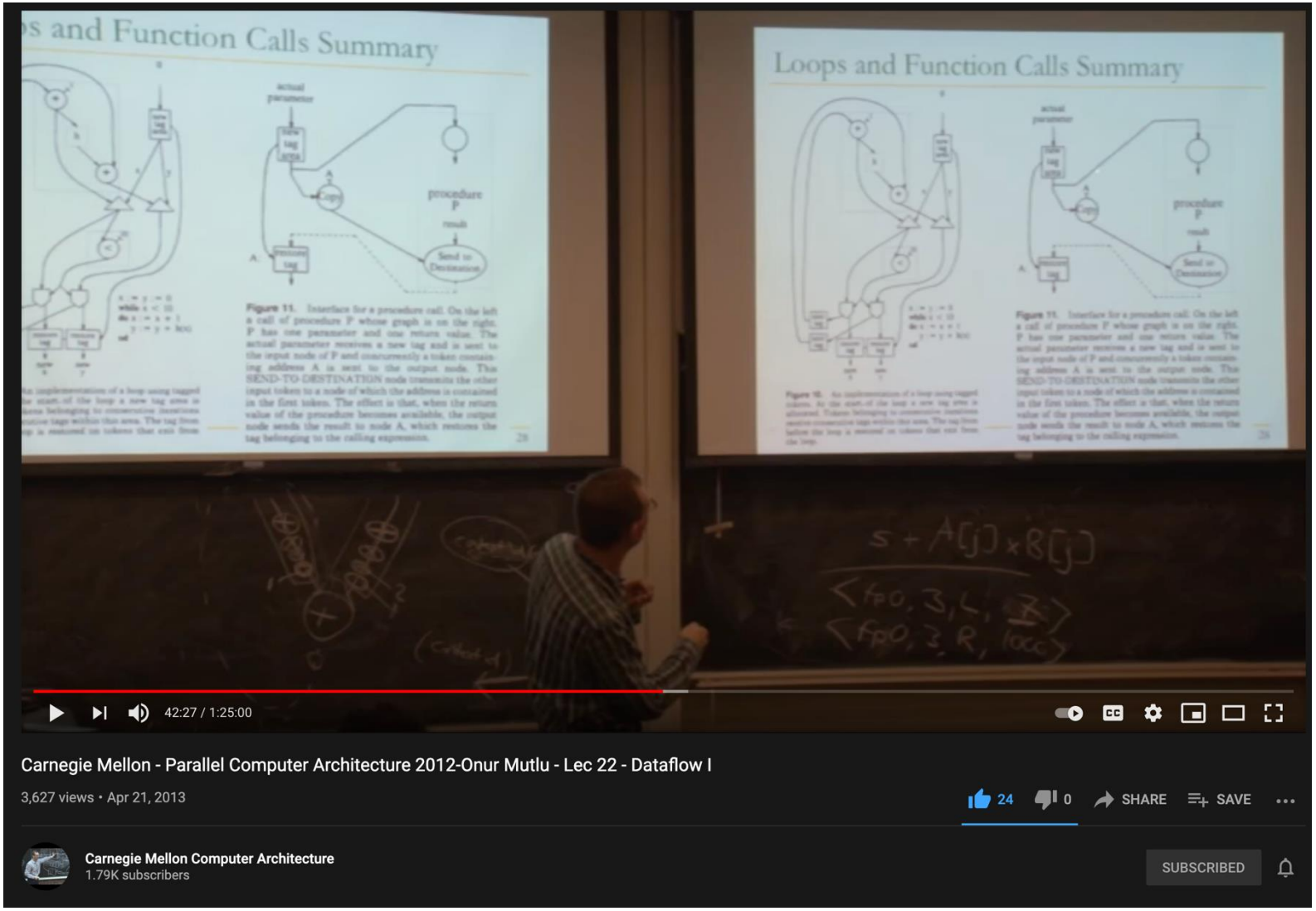
Let's Get Back to the von Neumann Model

- But, if you want to learn more about dataflow…

- Dennis and Misunas, "A preliminary architecture for a basic data-flow processor," ISCA 1974.
- Gurd et al., "The Manchester prototype dataflow computer," CACM 1985.
- A later lecture

- If you are really impatient:
  - http://www.youtube.com/watch?v=D2uue7izU2c
  - http://www.ece.cmu.edu/~ece740/f13/lib/exe/fetch.php?media=onur-740-fall13-module5.2.1-dataflow-part1.ppt

Lecture Video on Dataflow Architectures



http://www.youtube.com/watch?v=D2uue7izU2c

The von Neumann Model

- All major *instruction set architectures* today use this model
  - x86, ARM, MIPS, SPARC, Alpha, POWER, RISC-V, …
- Underneath (at the microarchitecture level), the execution model of almost all *implementations (or, microarchitectures)* is very different
  - Pipelined instruction execution: *Intel 80486 uarch*
  - Multiple instructions at a time: *Intel Pentium uarch*
  - Out-of-order execution: *Intel Pentium Pro uarch*
  - Separate instruction and data caches
- But, what happens underneath that is ***not* consistent** with the von Neumann model is ***not* exposed** to software
  - Difference between ISA and microarchitecture

What is Computer Architecture?

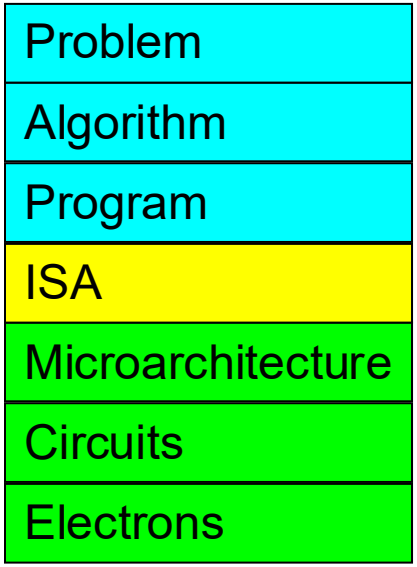
- **ISA+implementation definition:** The science and art of designing, selecting, and interconnecting hardware components and designing the hardware/software interface to create a computing system that meets functional, performance, energy consumption, cost, and other specific goals.

- **Traditional (ISA-only) definition:** "The term *architecture* is used here to describe the attributes of a system as seen by the programmer, i.e., the conceptual structure and functional behavior ***as distinct from*** the organization of the dataflow and controls, the logic design, and the physical implementation."

  Gene Amdahl, IBM Journal of R&D, April 1964

ISA vs. Microarchitecture

- ISA
  - Agreed upon interface between software and hardware
    - SW/compiler assumes, HW promises
  - What the software writer needs to know to write and debug system/user programs
- Microarchitecture
  - Specific implementation of an ISA
  - Not visible to the software
- Microprocessor
  - **ISA, uarch**, circuits
  - "Architecture" = ISA + microarchitecture



Microarchitecture

- A specific **implementation** of the ISA

- How do we implement the ISA?
  - We will discuss this for many lectures

- There can be many implementations of the same ISA
  - **MIPS** R2000, R3000, R4000, R6000, R8000, R10000, …
  - **x86**: Intel 80486, Pentium, Pentium Pro, Pentium 4, Kaby Lake, Coffee Lake, Comet Lake, Ice Lake, Golden Cove, Sapphire Rapids, …, AMD K5, K7, K9, Bulldozer, BobCat, Ryzen X, …
  - **POWER** 4, 5, 6, 7, 8, 9, 10 (IBM), …, **PowerPC** 604, 605, 620, …
  - **ARM** Cortex-M*, ARM Cortex-A*, NVIDIA Denver, Apple A*, M1, …
  - **Alpha** 21064, 21164, 21264, 21364, …
  - **RISC-V** …
  - …

ISA vs. Microarchitecture

- What is part of ISA vs. Uarch?
  - Gas pedal: interface for "acceleration"
  - Internals of the engine: implement "acceleration"

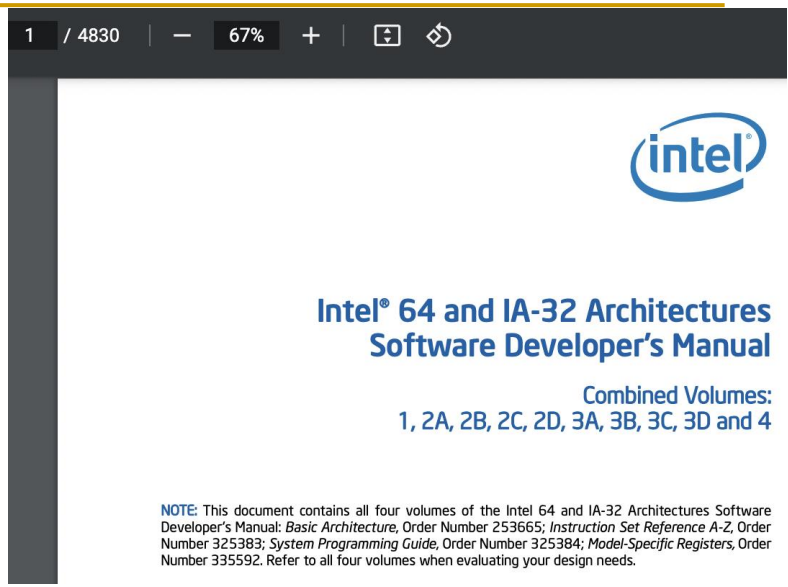
- Implementation (uarch) can be various as long as it satisfies the specification (ISA)
  - Add instruction vs. Adder implementation
    - Bit serial, ripple carry, carry lookahead adders are all part of microarchitecture **(see H&H Chapter 5.2.1)**
  - x86 ISA has many implementations:
    - Intel 80486, Pentium, Pentium Pro, Pentium 4, Kaby Lake, Coffee Lake, Comet Lake, Ice Lake, Golden Cover, Sapphire Rapids, …, AMD K5, K7, K9, Bulldozer, BobCat, Ryzen X, …

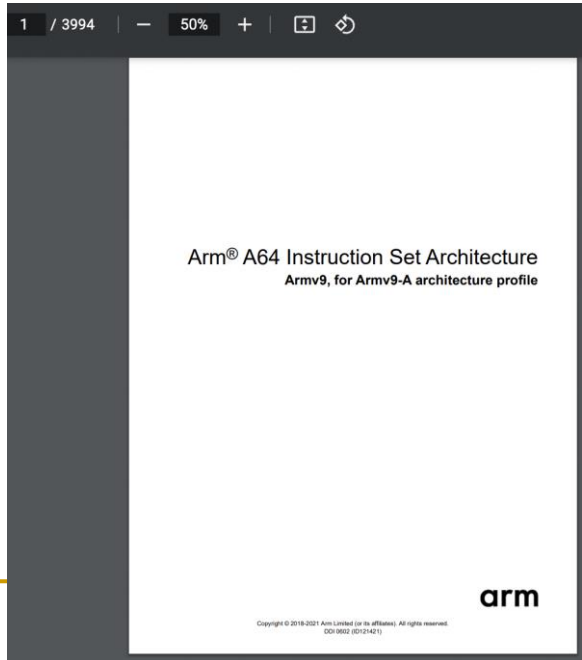
- Microarchitecture usually changes faster than ISA
  - Few ISAs (x86, ARM, SPARC, MIPS, Alpha, RISC-V) but many uarchs
  - *Why?*

https://www.vox.com/2015/7/1/8877583/two-foot-driving-pedal-error

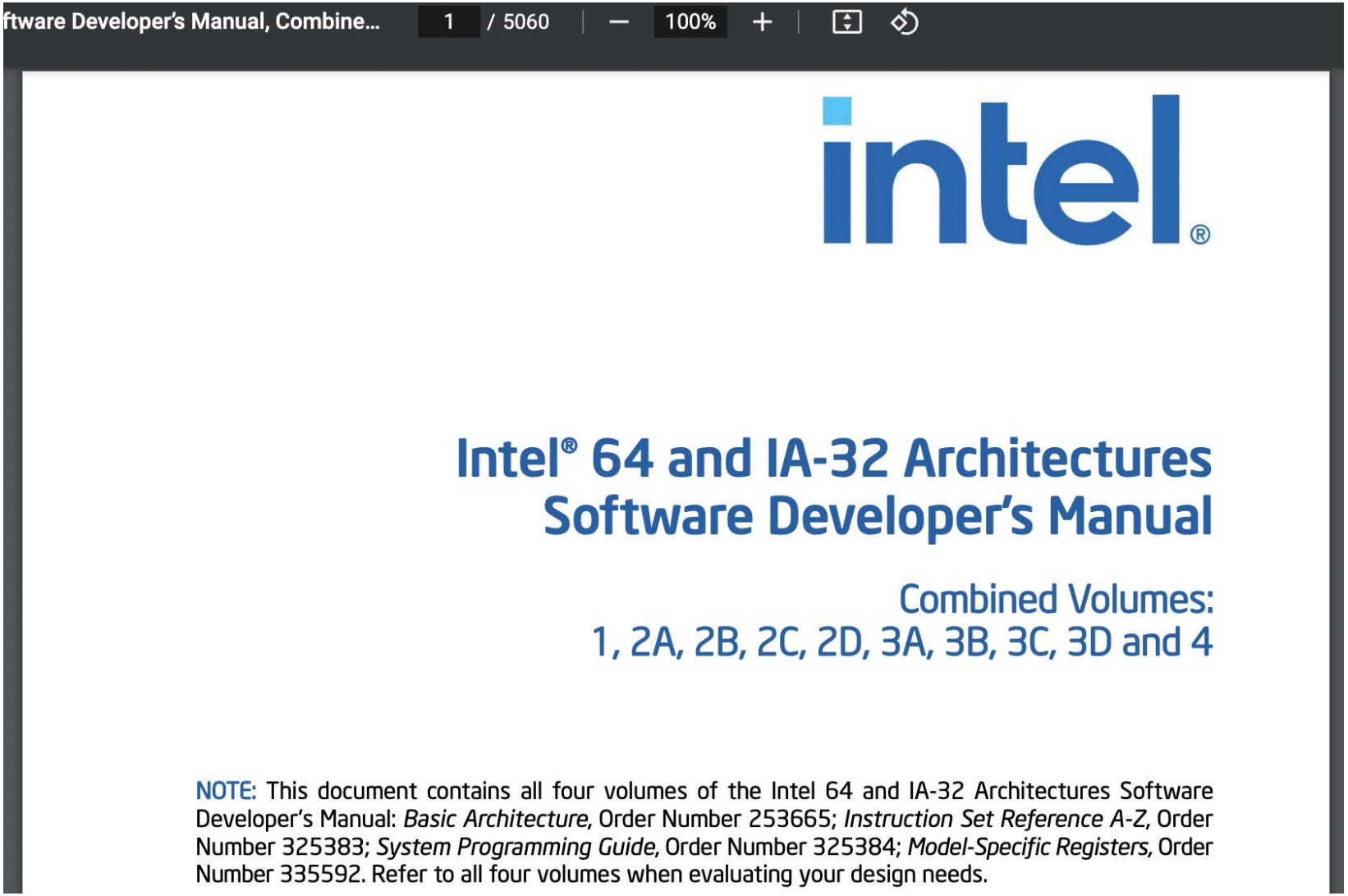
ISA: What Does It Specify?

- Instructions
  - Opcodes, Addressing Modes, Data Types
  - Instruction Types and Formats
  - Registers, Condition Codes
- Memory
  - Address space, Addressability, Alignment
  - Virtual memory management
- Call, Interrupt/Exception Handling
- Access Control, Priority/Privilege
- I/O: memory-mapped vs. instructions
- Task/thread Management
- Power & Thermal Management
- Multithreading & Multiprocessor support
- …



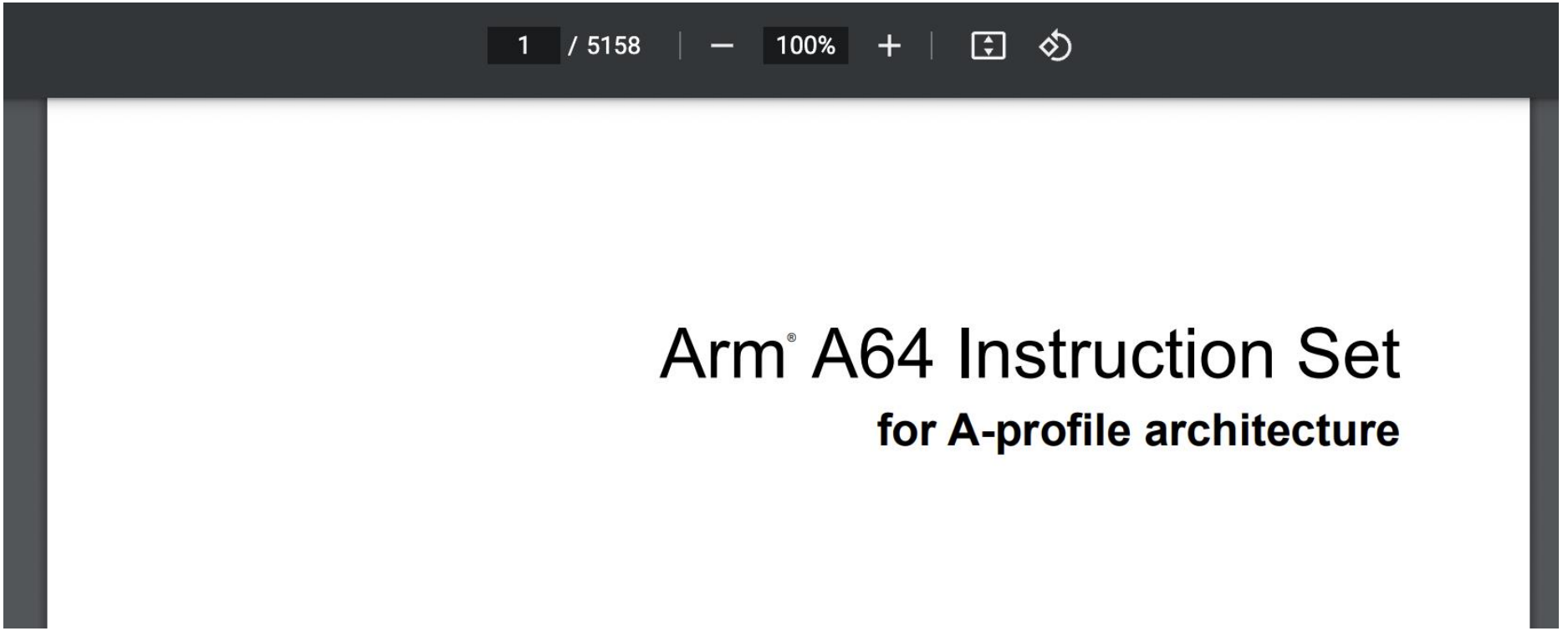


ISAs Keep Getting Extended



https://www.intel.com/content/www/us/en/developer/articles/technical/intel-sdm.html

ISAs Keep Getting Extended



https://developer.arm.com/documentation/ddi0602/latest/

ISA Manuals: Some Good Bedtime Reading

Combined Volume Set of Intel® 64 and IA-32 Architectures Software Developer's Manuals

Document	Description
Intel® 64 and IA-32 Architectures Software Developer's Manual Combined Volumes: 1, 2A, 2B, 2C, 2D, 3A, 3B, 3C, 3D, and 4	This document contains the following: **Volume 1**: Describes the architecture and programming environment of processors supporting IA-32 and Intel® 64 architectures. **Volume 2**: Includes the full instruction set reference, A-Z. Describes the format of the instruction and provides reference pages for instructions. **Volume 3**: Includes the full system programming guide, parts 1, 2, 3, and 4. Describes the operating-system support environment of Intel® 64 and IA-32 architectures, including: memory management, protection, task management, interrupt and exception handling, multi-processor support, thermal and power management features, debugging, performance monitoring, system management mode, virtual machine extensions (VMX) instructions, Intel® Virtualization Technology (Intel® VT), and Intel® Software Guard Extensions (Intel® SGX). NOTE: Performance monitoring events can be found here: https://perfmon-events.intel.com/ **Volume 4**: Describes the model-specific registers of processors supporting IA-32 and Intel® 64 architectures.
Intel® 64 and IA-32 Architectures Software Developer's Manual Documentation Changes	Describes bug fixes made to the Intel® 64 and IA-32 architectures software developer's manual between versions. NOTE: This change document applies to all Intel® 64 and IA-32 architectures software developer's manual sets (combined volume set, 4 volume set, and 10 volume set).

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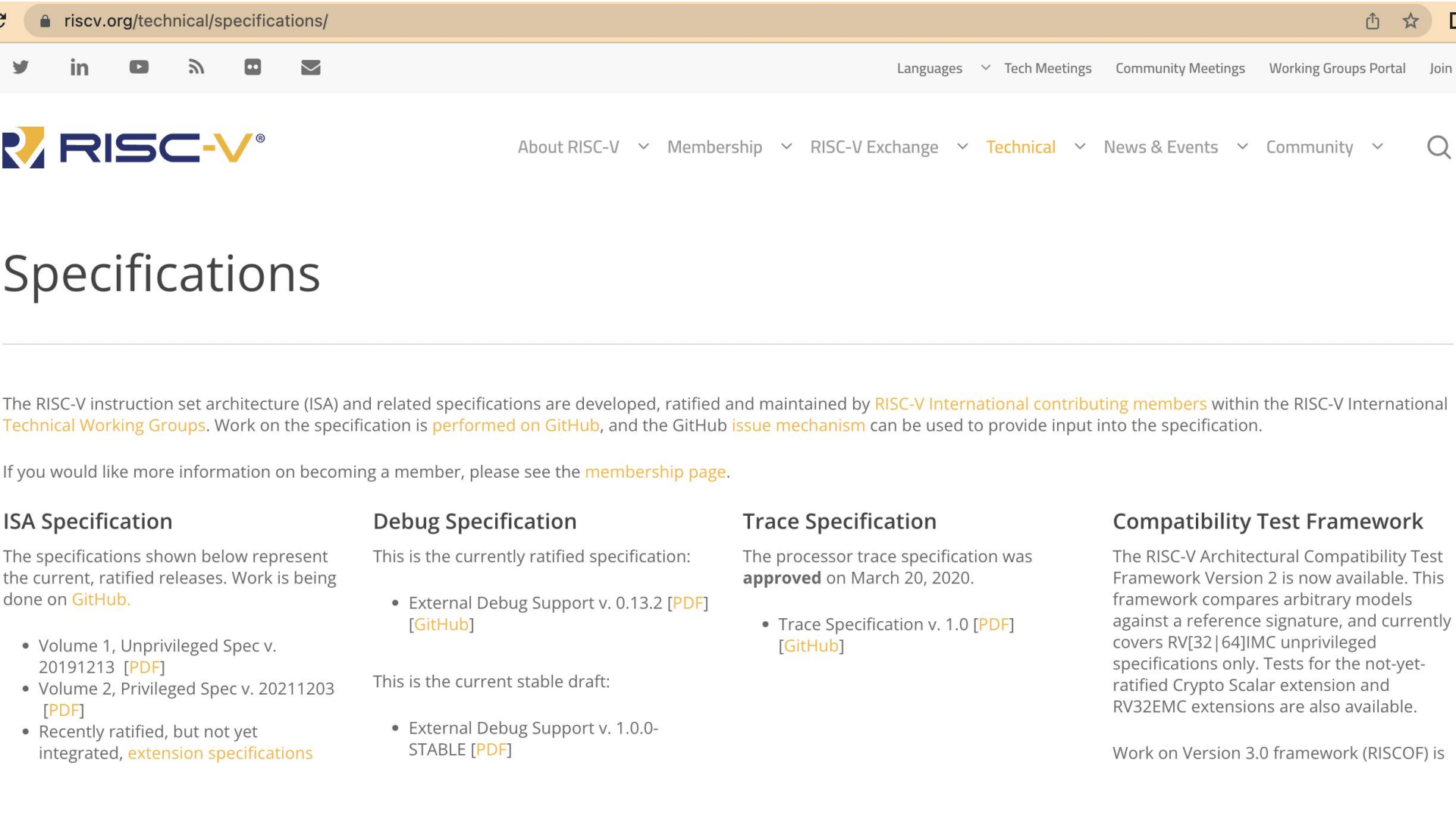
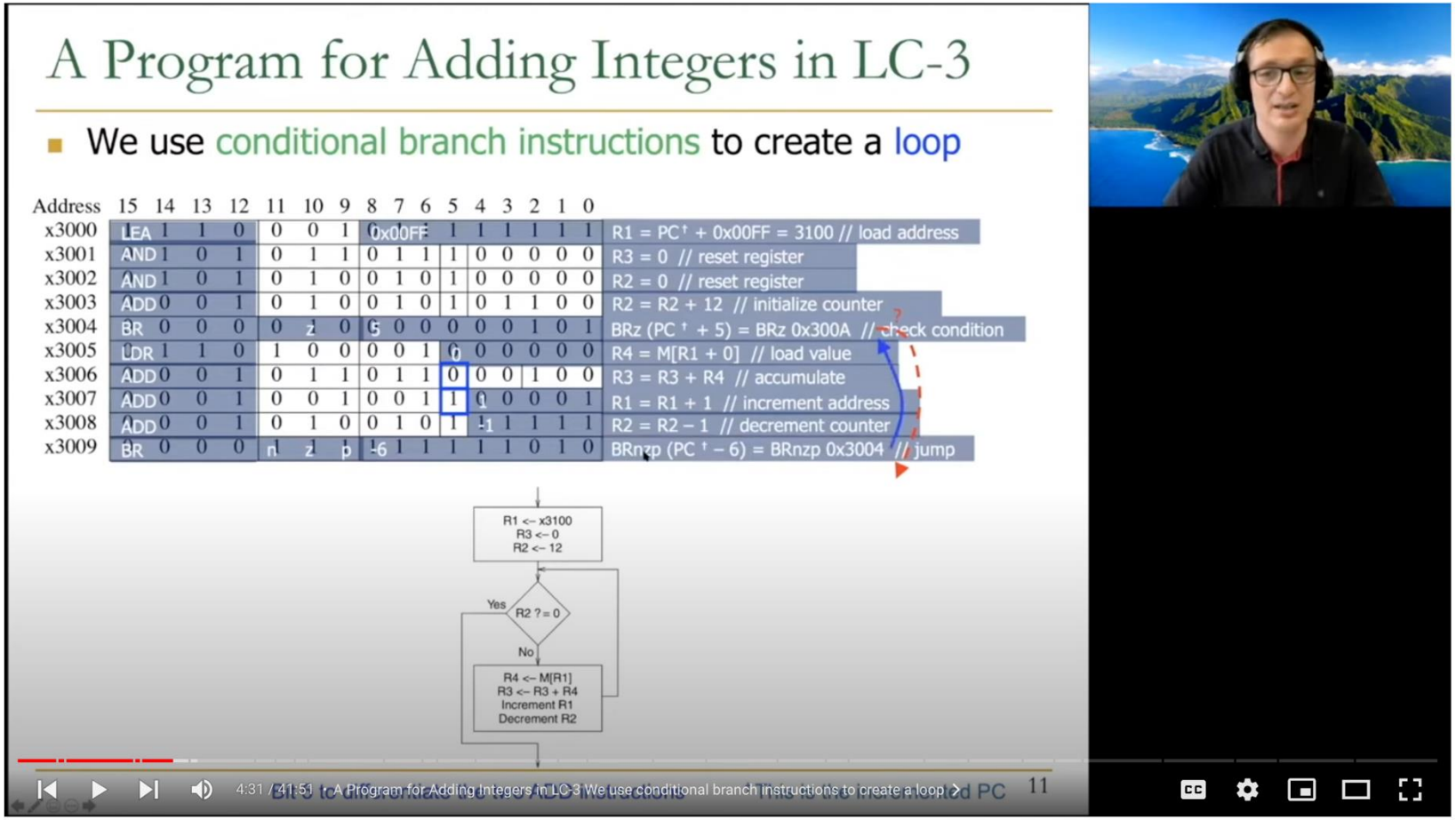


https://riscv.org/technical/specifications/

Lecture 9b: Assembly Programming



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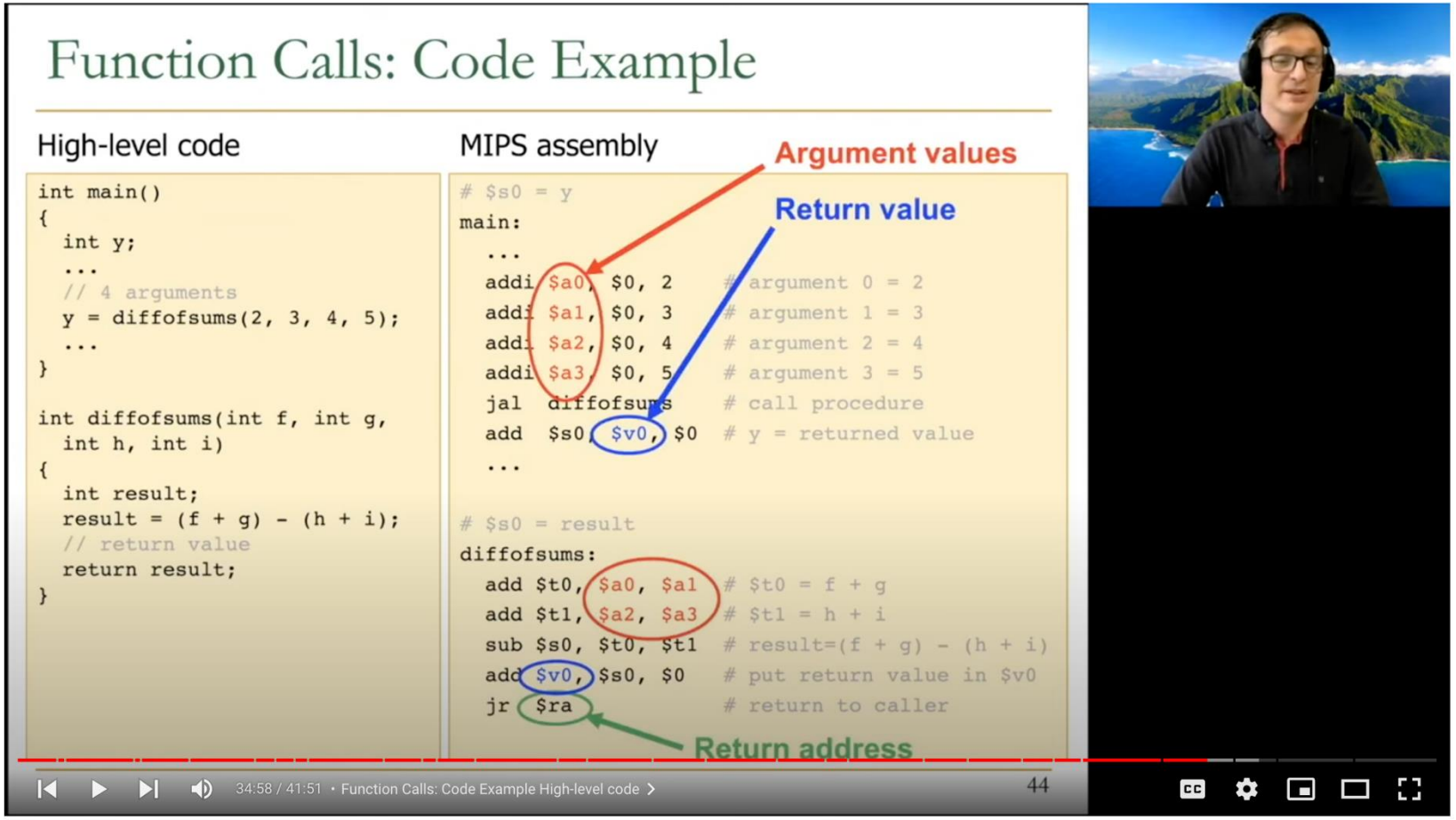

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